Phase-dependent core collection and window sensitivity

Verified computations, controls, numerical failures and remaining scientific questions

Purpose. This report separates reproducible numerical observations from invalid comparisons, numerical artifacts, known identities and untested scientific claims. It is a technical research report, not a claim of experimental discovery.

Evidence available. 9 frozen experiment batches were inspected. 19 hypotheses received numerical follow-up; 5 passed the performed mesh, two-window, longitudinal-sampling and independent-propagation agreement gates. Robustness is assessed separately and can fail even when a scalar observation is reproduced.

What was actually run. Monochromatic scalar angular-spectrum propagation, compared with independent real-space Fresnel quadrature. Source power, aperture, wavelength, observation window and the registered observable are retained for convergence tests. No model-generated executable code is run. Previously optimized controls are re-evaluated without refitting.

Interpretation. A nonzero scalar difference alone is not a new physical mechanism. Coherent interference, Gaussian identities and vortex parity are known effects. Negative predictions and failed numerical tests are retained. Numerical reproduction is not novelty verification, a full Maxwell calculation, an experimental result or a publication decision.

Reproduced observation: cycle_0006/C0006H001, signed effect 0.0192613; error floor 0.002; robust under all tested perturbations: True; registered prediction resolved with margin: True.

Reproduced observation: cycle_0006/C0006H003, signed effect -0.210269; error floor 0.002; robust under all tested perturbations: True; registered prediction resolved with margin: False.

Reproduced observation: cycle_0008/C0008H003, signed effect -0.192446; error floor 0.002; robust under all tested perturbations: True; registered prediction resolved with margin: False.

Reproduced observation: cycle_0009/C0009H001, signed effect -0.192977; error floor 0.002; robust under all tested perturbations: True; registered prediction resolved with margin: True.

Reproduced observation: cycle_0009/C0009H003, signed effect 0.00219489; error floor 0.002; robust under all tested perturbations: True; registered prediction resolved with margin: False.

Main findings and what they rule out

Airy window sensitivity. The original valley-filling ratio 1.603637 becomes 1.005804 under the first enlarged window at fine grid spacing. Thus an apparent 60.36% increase becomes 0.58% in that diagnostic. The original large effect does not survive this window change; it is not reliable evidence for a stable large enhancement. The ridge statistic and boundary handling require further controlled study.

Phase-dependent Pearcey core collection. At fixed source scale 150 micrometres and the frozen core/window, coherent versus incoherent combination gives the following differences in collected power fraction: phase 0 rad: +1.9261 percentage points; phase 3.14159 rad: -21.0269 percentage points. These are separate conditions, not an optimized phase search or a newly established physical law.

The frozen batch predicted a nonnegative difference for its registered cases. The reproducible negative difference rejects the registered pi-phase directional prediction; it does not reject the phase-zero primary prediction. It remains useful evidence for phase-sensitive redistribution, to be interpreted using the known interference cross term and equal-power normalization. The untested mechanism and prior-art question must be kept separate from numerical reproduction.

Comparison audit. Empty source transformations in a transformed-versus-base experiment retain the same environment on both arms, so they cannot test lens-on versus lens-off. The repaired interface provides an explicit lens-versus-free-space comparison and rejects that empty-transform design at online compilation. Exact odd-vortex cancellation is retained as an undefined equal-power comparison, not converted into a positive finding.

Research value and next use. These computations identify which scalar effects survive independent propagation and numerical refinement, reject a broad phase-sign prediction, expose a window-sensitive apparent enhancement, and specify a reproducible next experiment. They support a focused mechanism and robustness study; they do not establish a new beam family, an experimental result or novelty.

Measurement and validation contract

A source is normalized to 1 W within a fixed circular aperture of radius 0.9 mm. The original computational window is 2.4 mm; wavelength is 633 nm unless a held-out wavelength is explicitly identified. The Gaussian envelope waist is 300 micrometres. Beam scale has family-specific meaning: it does not change the Gaussian waist in the present field generator.

For the supplied Pearcey source, C4(u,v) is the canonical integral of exp[i(t^4 + v*t^2 + u*t)] over the real contour, evaluated by a convergent contour deformation, then multiplied by an explicit Gaussian envelope and circular aperture. The Butterfly source fixes two higher control parameters to zero. No polynomial phase mask is renamed as a canonical beam.

Core efficiency is the median over the frozen observation planes of power inside a fixed on-axis circular disk, divided by input power. Valley filling is a midpoint-to-ridge intensity statistic in a transverse marginal profile and requires matched qualifying ridge pairs. It is not a direct phase measurement. Brightest-pixel trajectories are ambiguous when there are degenerate maxima.

Fixed follow-up: 256 and 384 grids at 2.4 mm, windows of 3.6 and 4.8 mm at the fine grid spacing, doubled longitudinal intervals, and a separate real-space Fresnel calculation at the reference grid. Agreement is required both for the signed comparison effect and for the two actual arm observables. All solver, boundary and power gates must pass.

Tolerances fixed by policy v1: relative tolerance 2.0%; dimensionless absolute floor 0.002; ratio floor 0.01; position floor 3.125 micrometres. A signed prediction is resolved only if its margin exceeds the larger of the observed refinement discrepancy and the fixed floor. Meeting a weak >=0 inequality at zero is insufficient.

Separate robustness probes: 256-level phase quantization; a fixed 640 nm wavelength hold-out with residual optical-path scaling; two fixed 0.02 rad input phase-noise realizations. No parameters are optimized on these probes. These tests are limited diagnostics, not statistical confidence intervals. Random pixel-noise screens are not a continuous stochastic field coupled across grids; their mesh interpretation requires caution.

Controls: defocus, quadratic-plus-quartic and four-parameter Gaussian residual phases were optimized under equal original search budgets. The stored best parameters are re-evaluated at the reference grid under the same composition and environment as each candidate. This is not a proof of superiority over an exhaustive classical family.

Evidence batch cycle_0002

For a finite-energy Airy field paired with its physical-origin 180-degree rotated copy, does coherent rather than incoherent composition increase the registered valley_filling statistic over 0.04–0.08 m when two prominent ridges qualify?

Registered proposed mechanism: Coherent composition adds a phase-dependent cross term to the component intensities. For a rotated Airy pair, this term may redistribute intensity from the ridges into the central valley; the corresponding incoherent composition removes the cross term. Valley_filling tests the predicted intensity redistribution, not phase directly.

Observable: valley_filling. Comparison: coherent_vs_incoherent (coherent versus incoherent). Effect: ratio; frozen prediction: >= 1.0. A comparison across different source cases is not implicit in this interface.

Window: 0.04 to 0.08 m, 11 planes; core radius 100 micrometres. Source artifact: cycle_0002/frozen_protocol.json.

C0002H001: airy, scale 100 micrometres, source transformations []; copy angle 3.142, phase 0 rad, amplitude weight 1; environment free_space. Original effect not defined, testable False; failed basic checks: window_edges.

C0002H002: airy, scale 100 micrometres, source transformations []; copy angle 3.142, phase 1.571 rad, amplitude weight 1; environment free_space. Original effect 1.60364, testable True; failed basic checks: window_edges.

C0002H003: airy, scale 100 micrometres, source transformations []; copy angle 3.142, phase 3.142 rad, amplitude weight 1; environment free_space. Original effect 1.30936, testable True; failed basic checks: window_edges.

Follow-up routing: C0002H001 -> completed; unconfirmed.

Follow-up routing: C0002H002 -> completed; unconfirmed.

Follow-up routing: C0002H003 -> completed; unconfirmed.

C0002H001: unconfirmed; reference effect not defined; numerical error floor 0.01. Failed confirmation gates: mesh_convergence, window_convergence, second_window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes: quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: True.

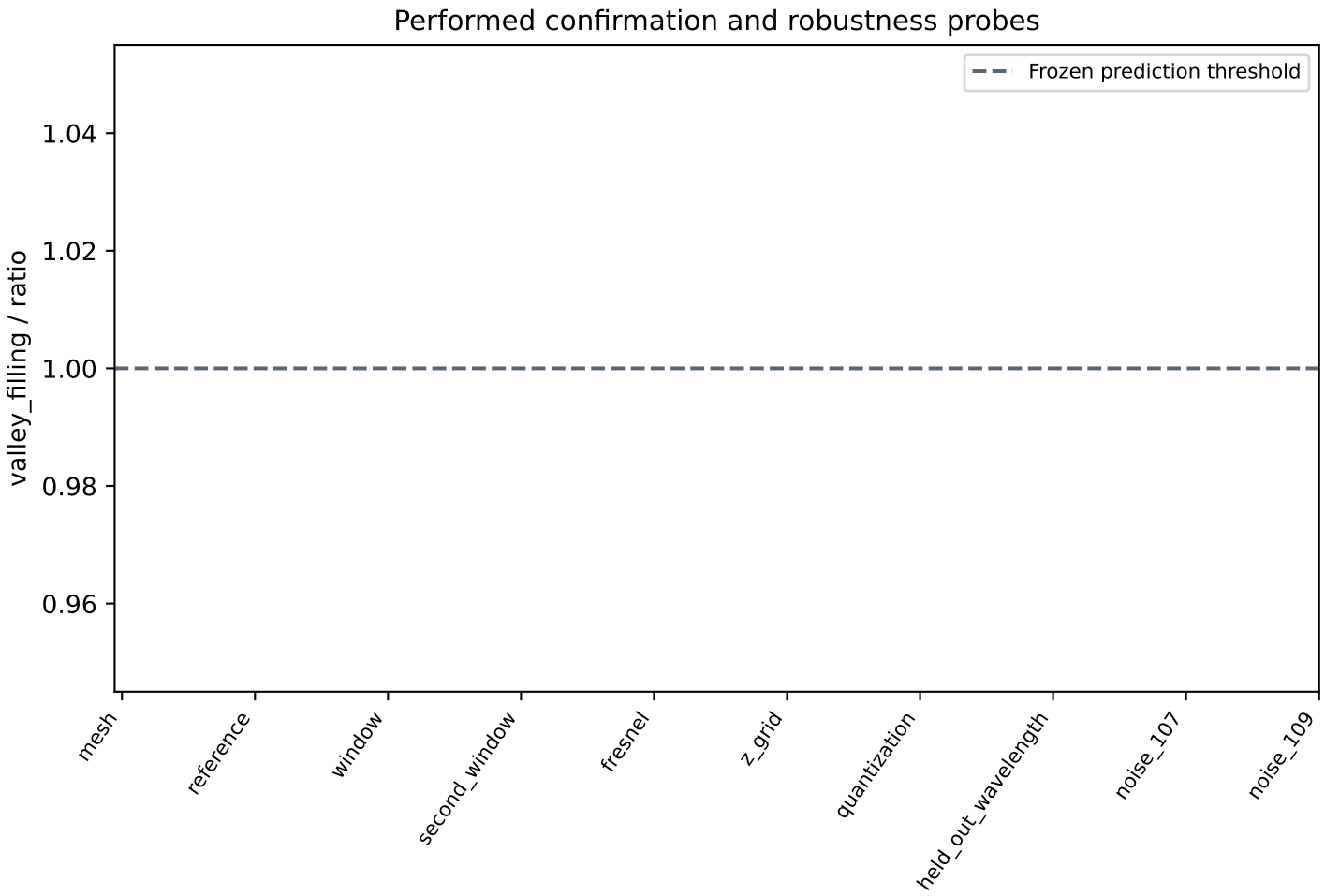
C0002H002: unconfirmed; reference effect 1.46974; numerical error floor 0.493625. Failed confirmation gates: mesh_convergence, window_convergence, second_window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes:

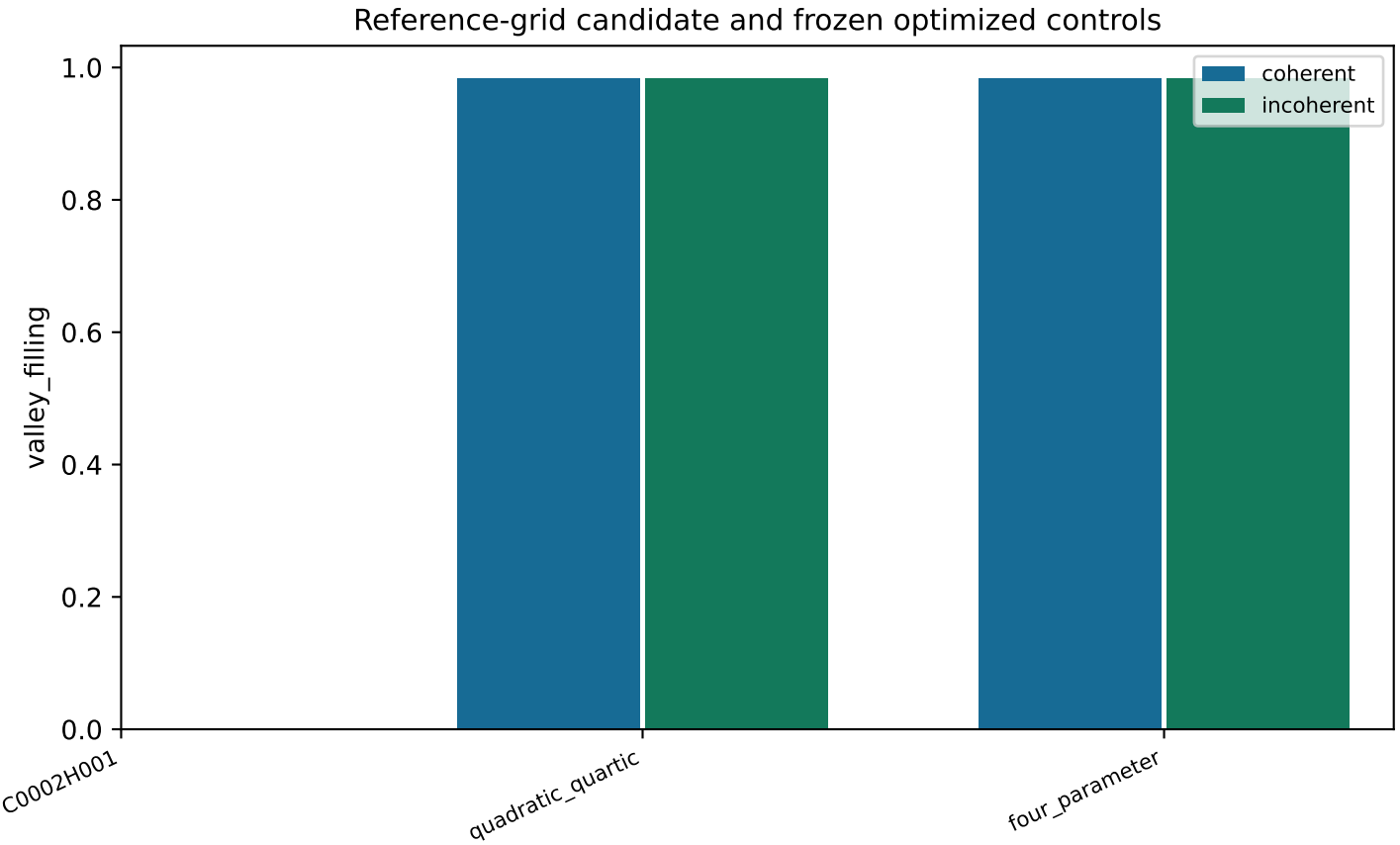
Evidence batch cycle_0002 (continued)

quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: True.

C0002H003: unconfirmed; reference effect 1.21357; numerical error floor 0.407066. Failed confirmation gates: mesh_convergence, window_convergence, second_window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes: quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: False.

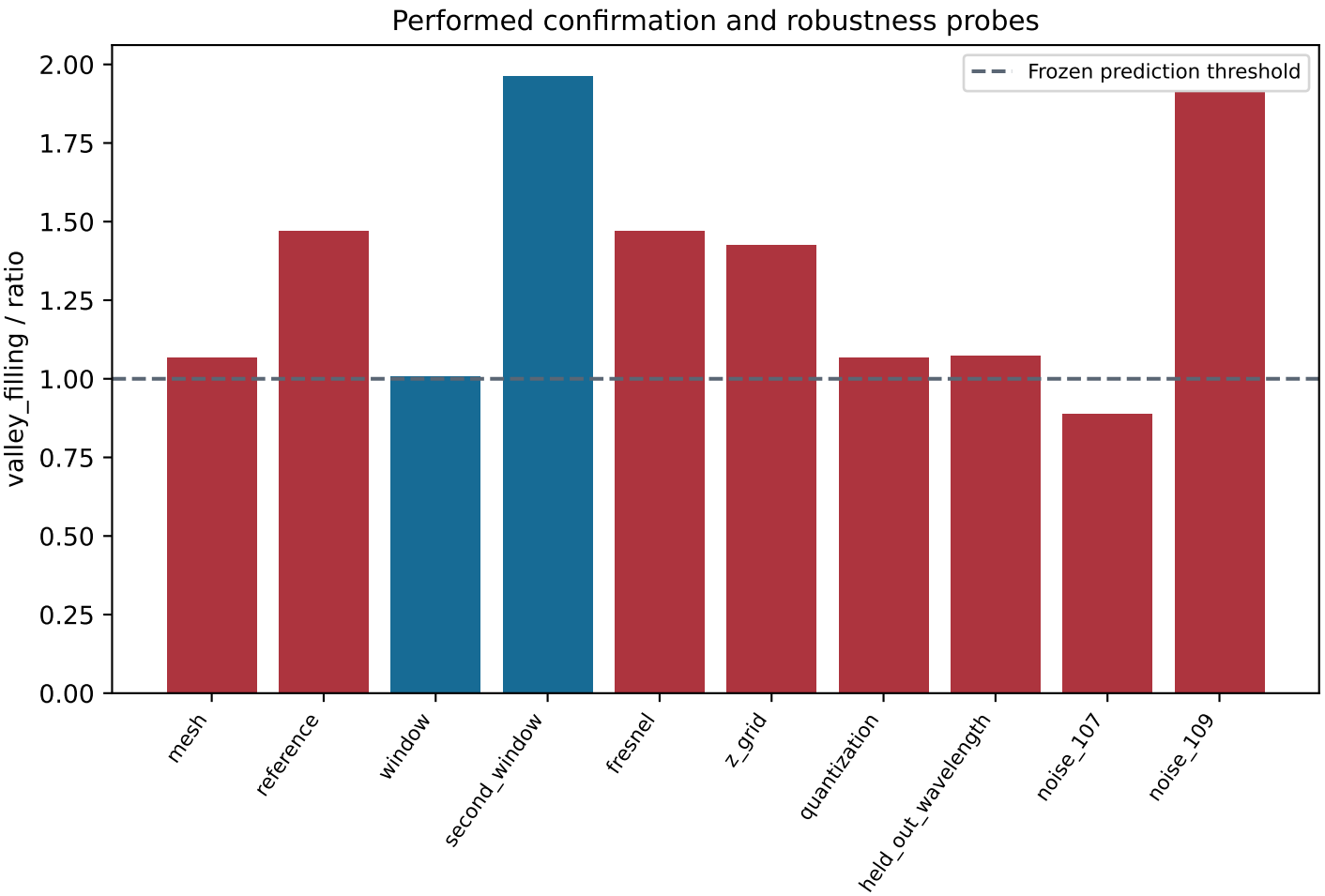
cycle_0002 / C0002H001

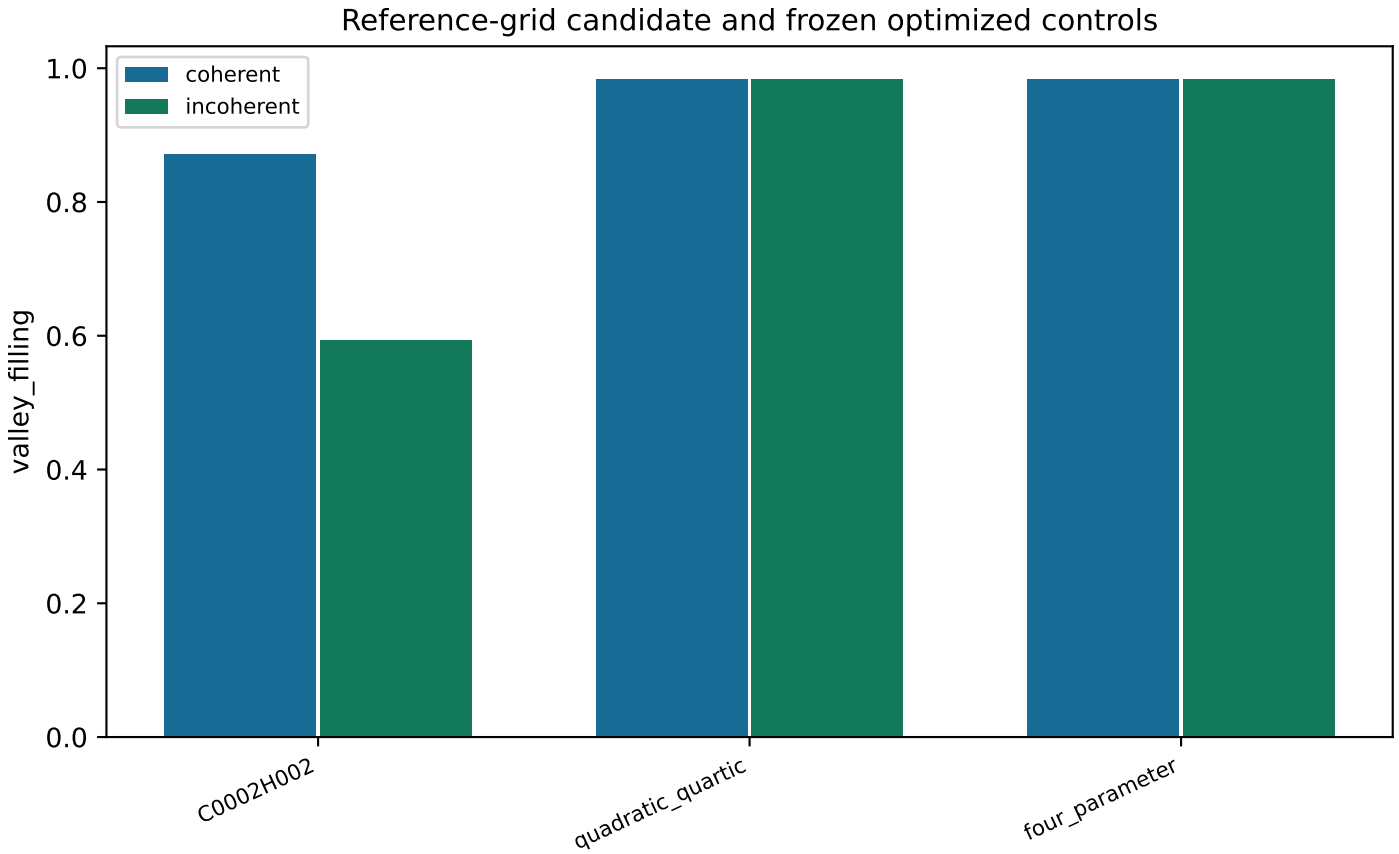




Red bars fail at least one numerical gate. Missing measurements are not zero.
Controls are re-evaluations of frozen parameters, not exhaustive optima or a novelty test.

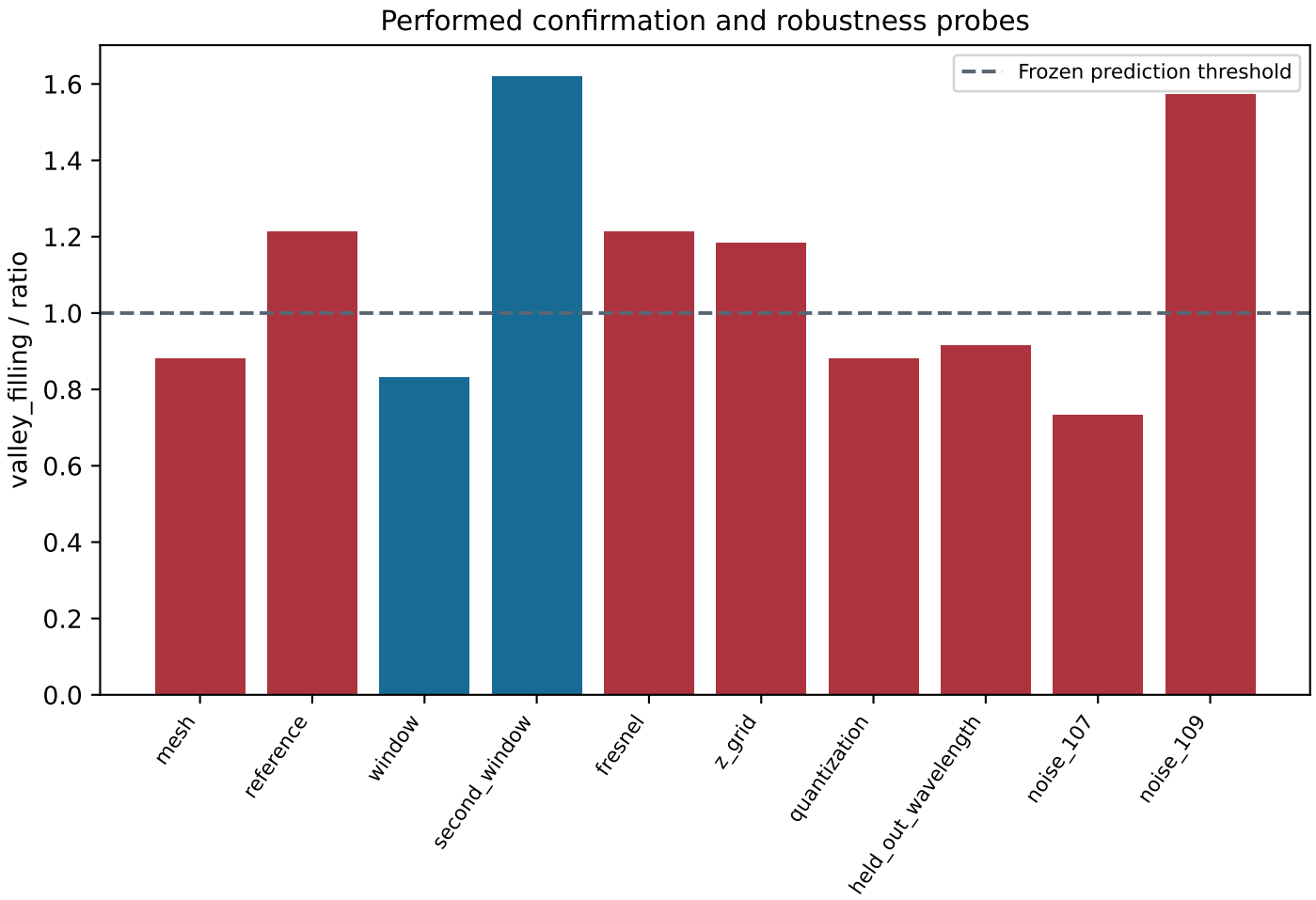
cycle_0002 / C0002H002

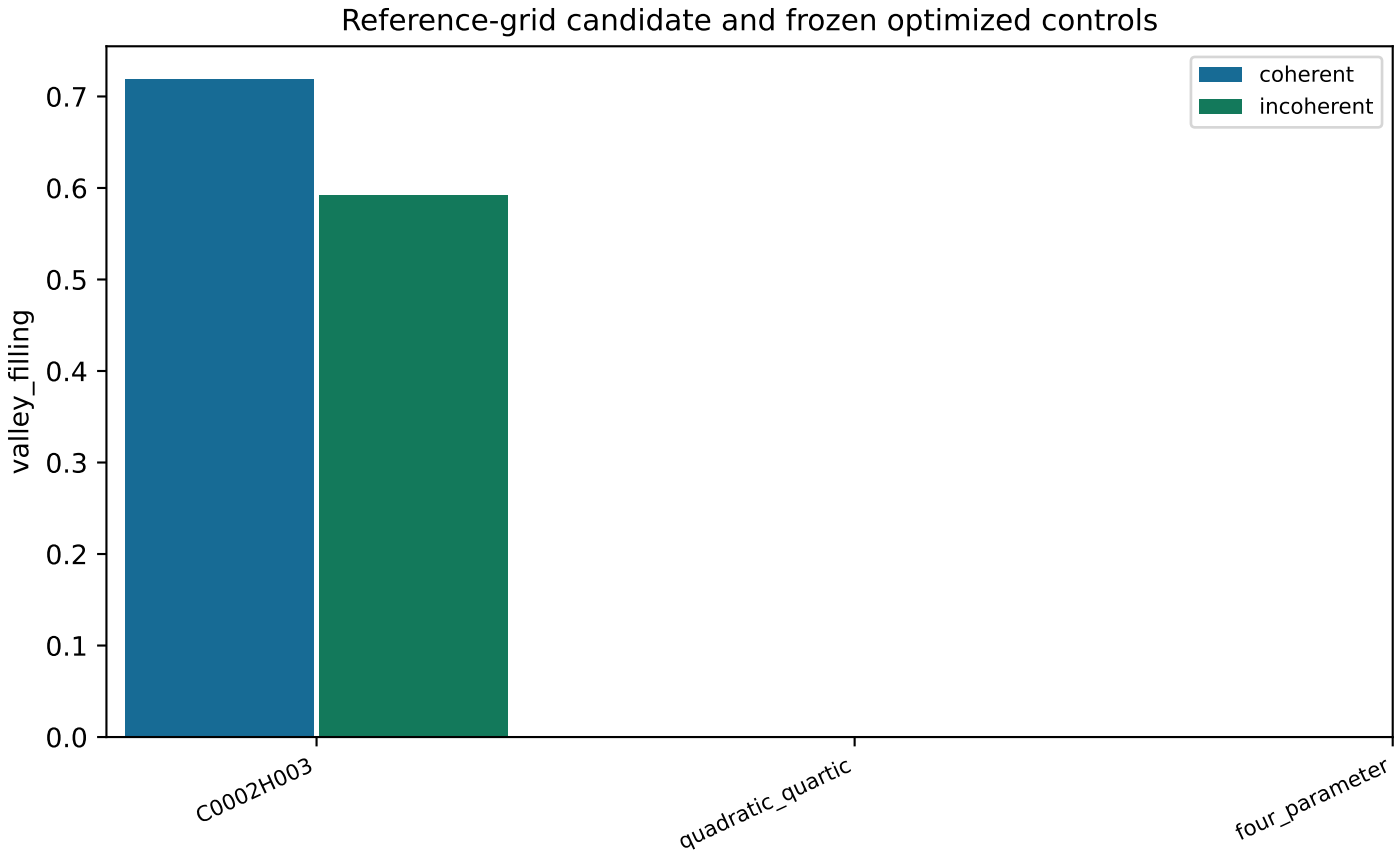




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cycle_0002 / C0002H003





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Evidence batch cycle_0005

For a fixed finite-energy Airy input and a preregistered free-space observation window, does adding a quadratic phase chirp increase peak_bending_m relative to the same unchirped base field?

Registered proposed mechanism: A quadratic input phase changes the transverse phase gradient and propagation directions, potentially altering how the brightest-pixel trajectory bends relative to its endpoint chord. The predicted direction is a test in each registered chirp regime, not a universal property.

Observable: peak_bending_m. Comparison: transformed_vs_base (single versus base). Effect: difference; frozen prediction: >= 0. A comparison across different source cases is not implicit in this interface.

Window: 0.04 to 0.08 m, 11 planes; core radius 110 micrometres. Source artifact: cycle_0005/frozen_protocol.json.

C0005H001: airy, scale 100 micrometres, source transformations [{'kind': 'chirp', 'amount': 0.25}]; copy angle 3.142, phase 0 rad, amplitude weight 1; environment free_space. Original effect 1.36069e-06, testable True; failed basic checks: window_edges.

C0005H002: airy, scale 100 micrometres, source transformations [{'kind': 'chirp', 'amount': 0.6}]; copy angle 3.142, phase 0 rad, amplitude weight 1; environment free_space. Original effect 2.65165e-06, testable True; failed basic checks: window_edges.

C0005H003: airy, scale 100 micrometres, source transformations [{'kind': 'chirp', 'amount': -0.25}]; copy angle 3.142, phase 0 rad, amplitude weight 1; environment free_space. Original effect -5.3033e-06, testable True; failed basic checks: window_edges.

Follow-up routing: C0005H001 -> completed; unconfirmed.

Follow-up routing: C0005H002 -> completed; unconfirmed.

Follow-up routing: C0005H003 -> completed; unconfirmed.

C0005H001: unconfirmed; reference effect -8.83883e-06; numerical error floor 5.14926e-06. Failed confirmation gates: mesh_convergence, window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes: quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: True.

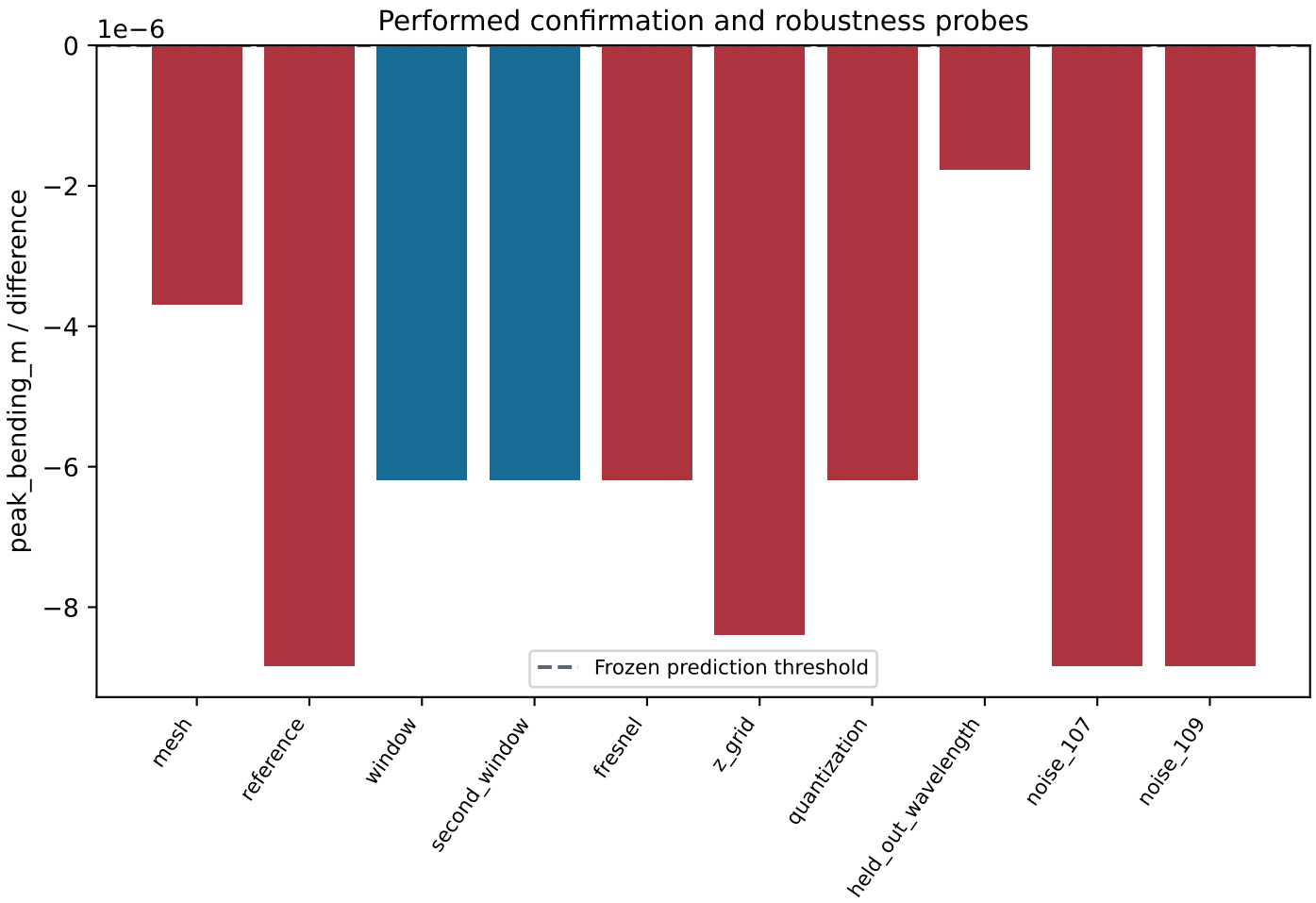
C0005H002: unconfirmed; reference effect 3.53553e-06; numerical error floor 4.41942e-06. Failed confirmation gates: mesh_convergence, window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes: quantization,

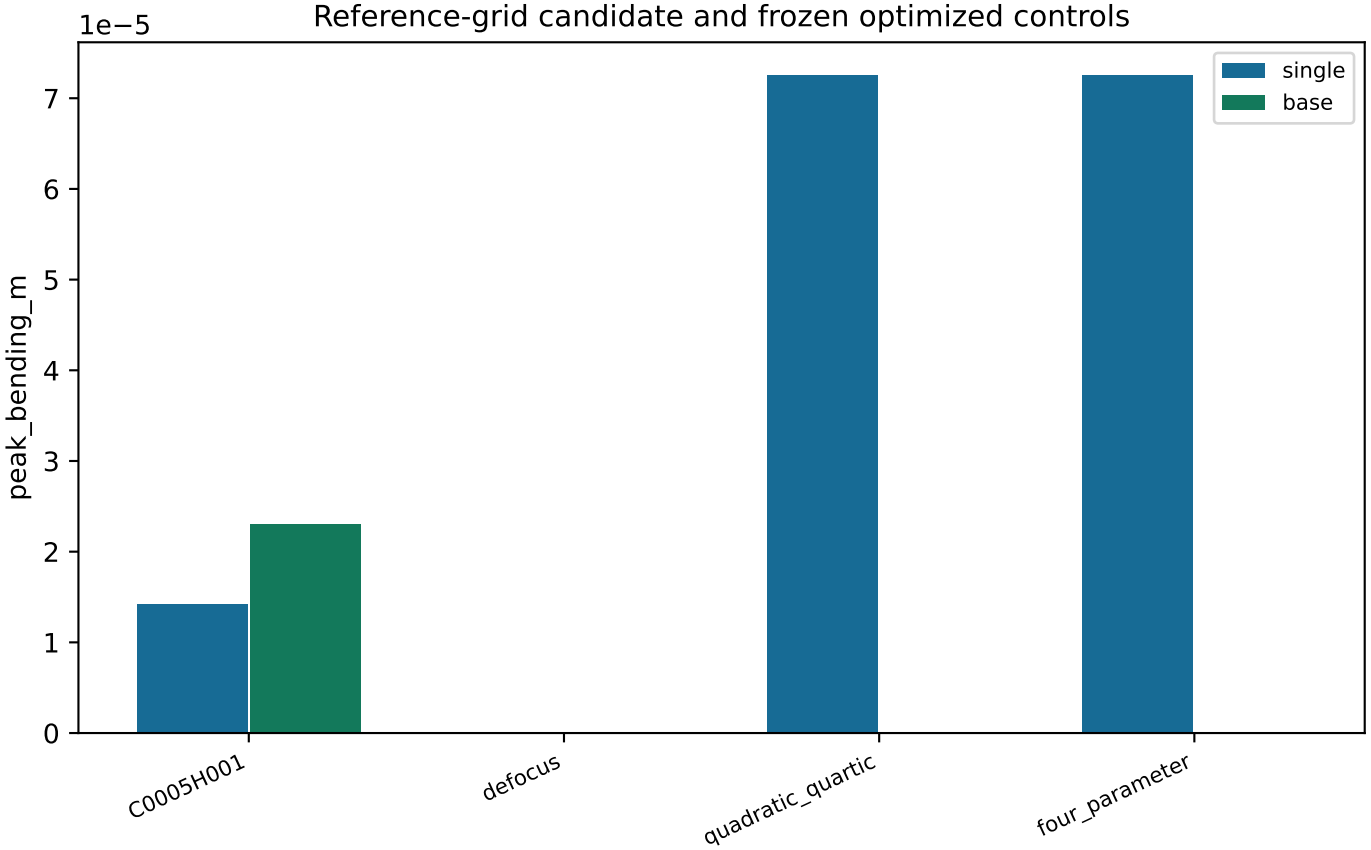
Evidence batch cycle_0005 (continued)

held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: True.

C0005H003: unconfirmed; reference effect -4.75925e-06; numerical error floor 4.41942e-06. Failed confirmation gates: mesh_convergence, window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes: quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: True.

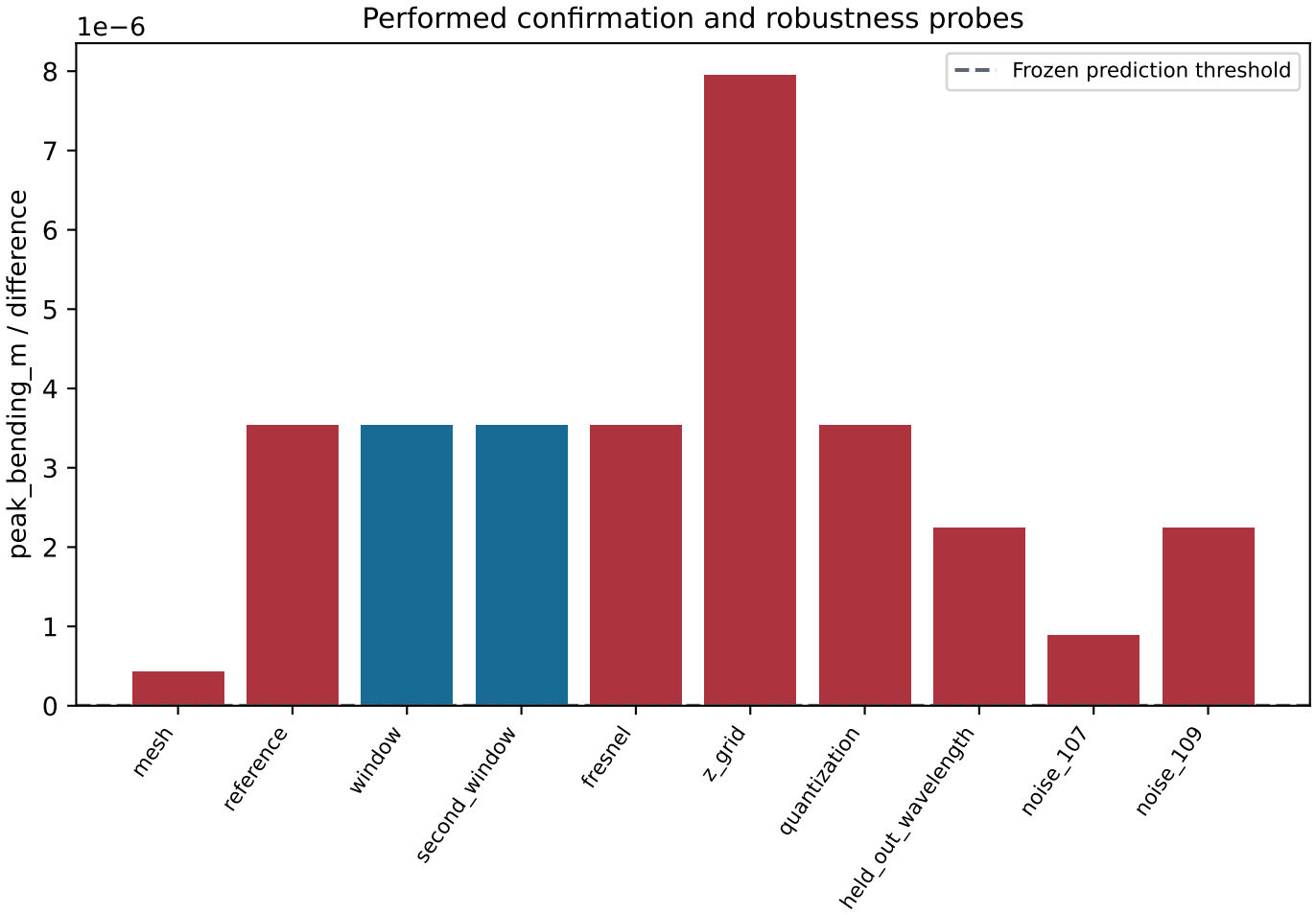
cycle_0005 / C0005H001

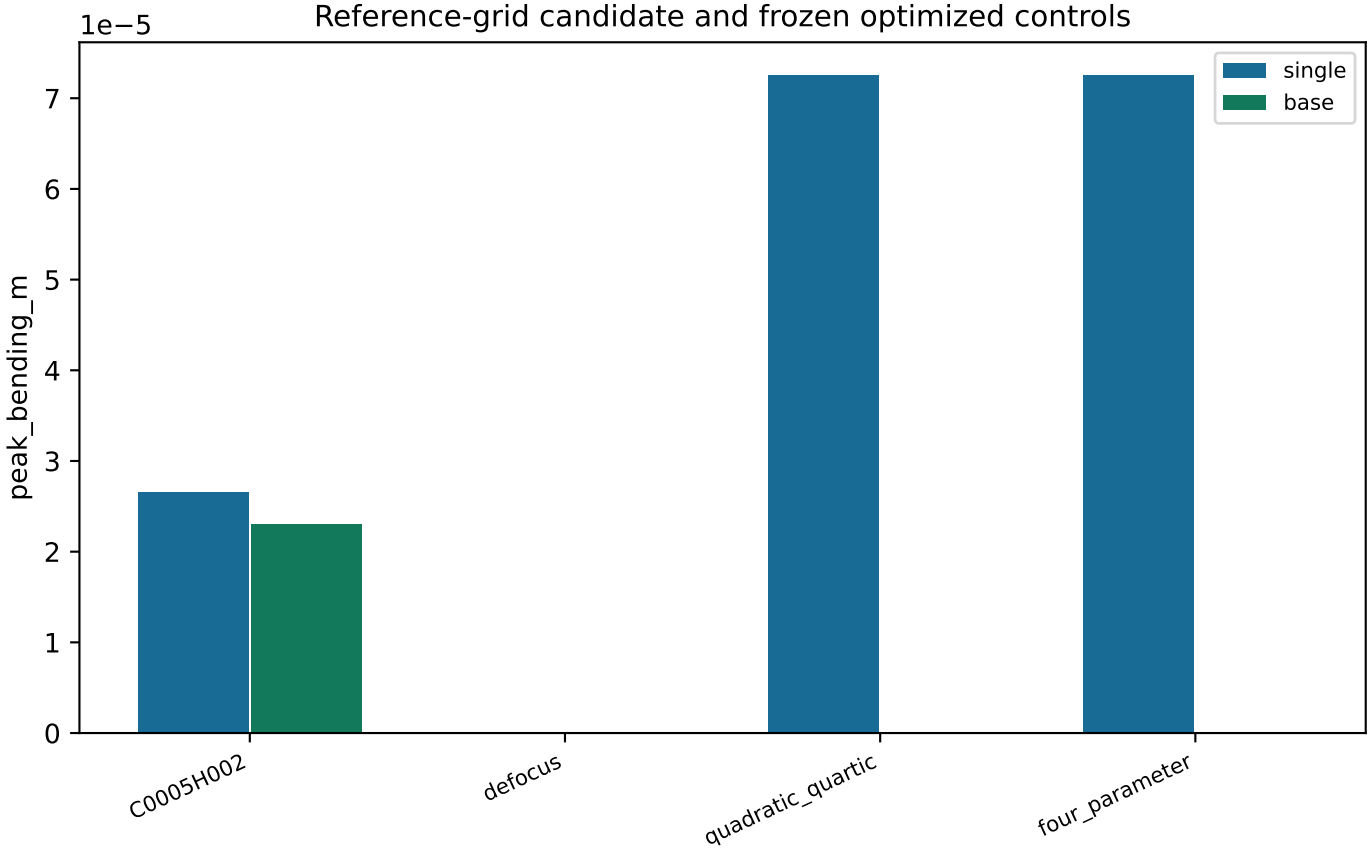




Red bars fail at least one numerical gate. Missing measurements are not zero.
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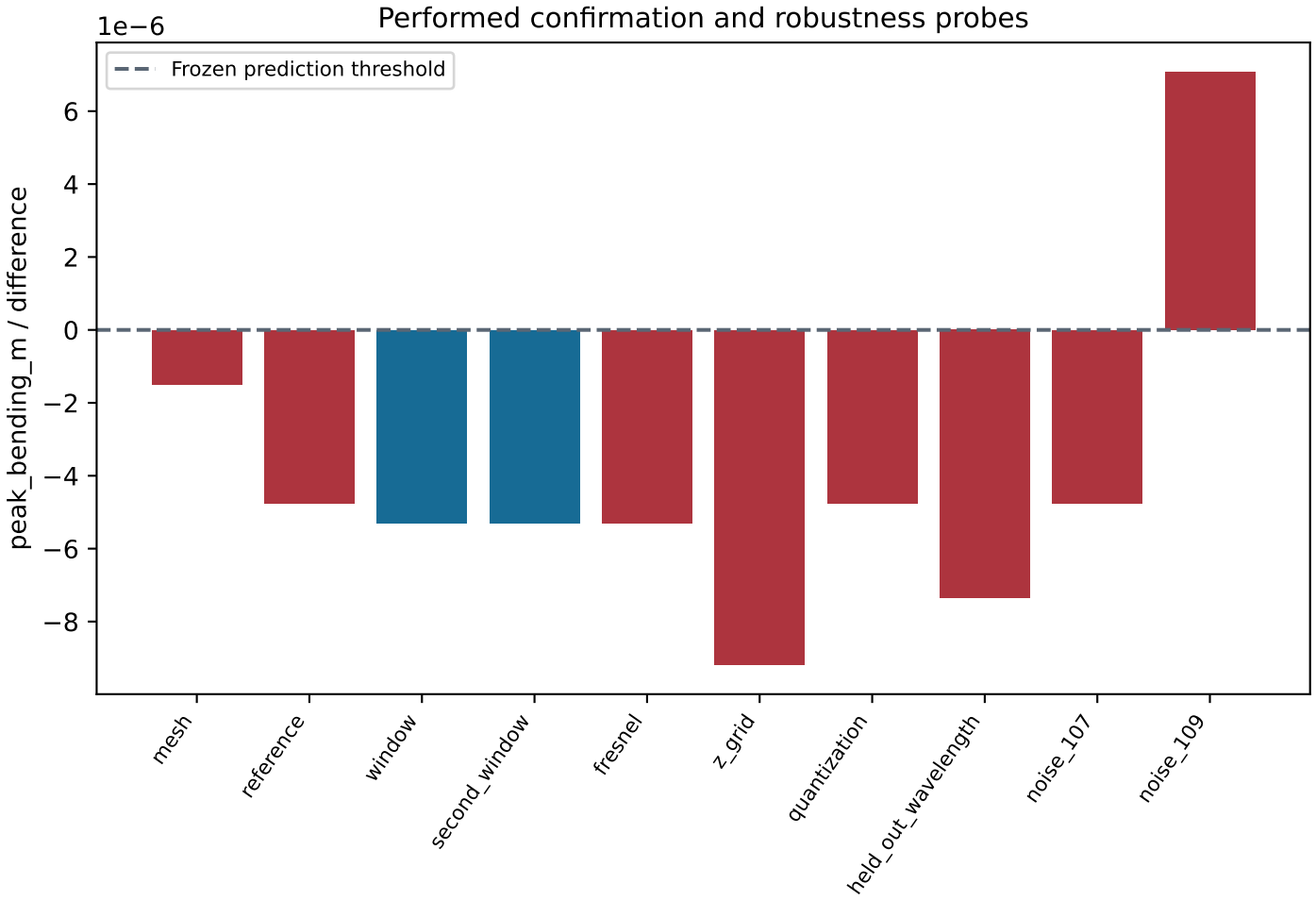
cycle_0005 / C0005H002

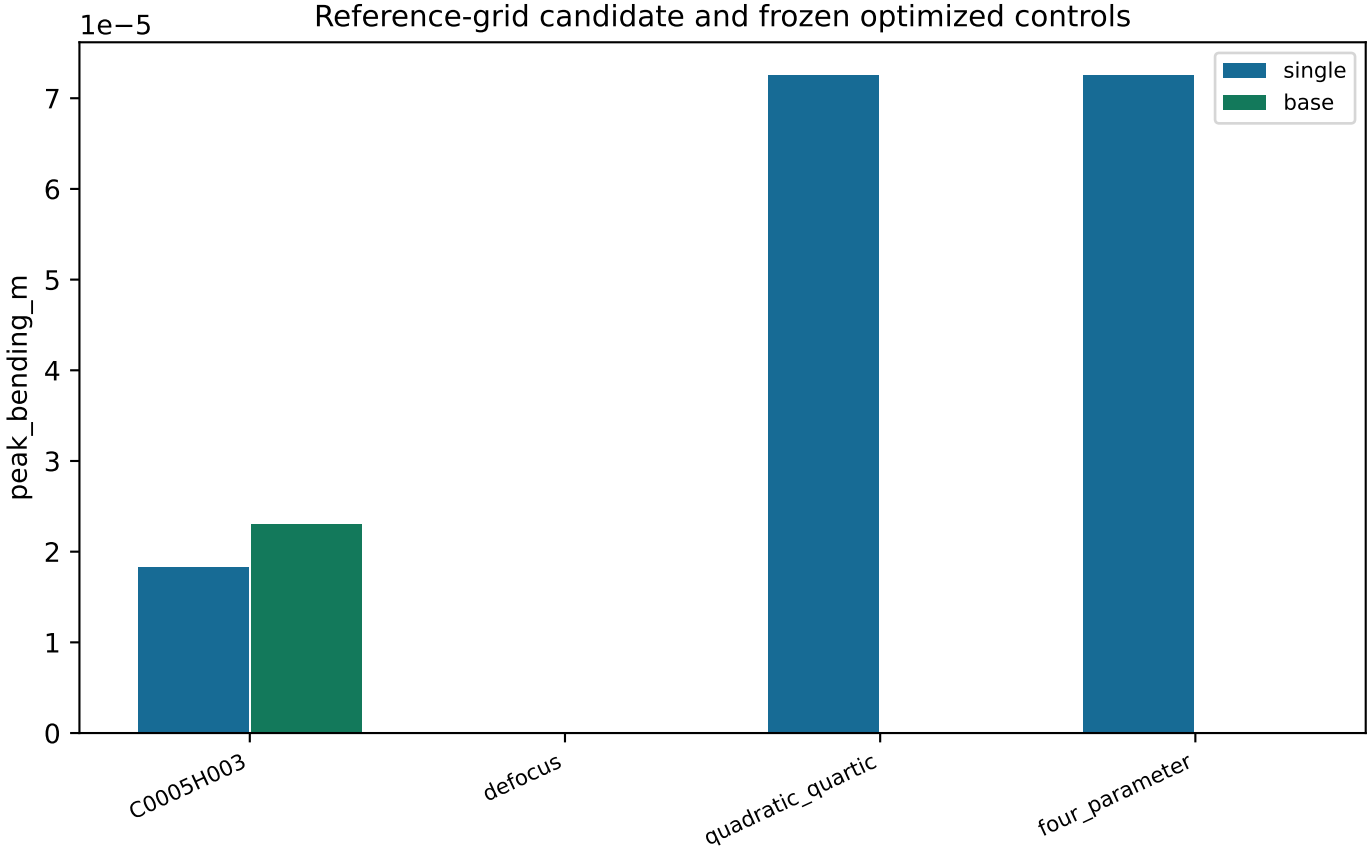




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cycle_0005 / C0005H003





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Evidence batch cycle_0006

For a fixed apodized canonical Pearcey input and its equal-power 180-degree-rotated copy, does coherent superposition at relative phase zero produce greater median on-axis core efficiency than the corresponding incoherent pair over 0.04–0.08 m in free space? The phase-zero case is primary; the phase-pi/2 and phase-pi cases are preregistered phase-dependence controls.

Registered proposed mechanism: Coherent addition introduces a relative-phase-dependent interference cross term between the Pearcey field and its physical-origin 180-degree-rotated copy. This term can redistribute power within the fixed on-axis circular core; comparison with the same incoherent arms isolates the coherence-dependent contribution.

Observable: core_efficiency. Comparison: coherent_vs_incoherent (coherent versus incoherent). Effect: difference; frozen prediction: >= 0. A comparison across different source cases is not implicit in this interface.

Window: 0.04 to 0.08 m, 9 planes; core radius 100 micrometres. Source artifact: cycle_0006/frozen_protocol.json.

C0006H001: pearcey, scale 150 micrometres, source transformations []; copy angle 3.142, phase 0 rad, amplitude weight 1; environment free_space. Original effect 0.0193172, testable True; failed basic checks: none.

C0006H002: pearcey, scale 150 micrometres, source transformations []; copy angle 3.142, phase 1.571 rad, amplitude weight 1; environment free_space. Original effect 0, testable True; failed basic checks: none.

C0006H003: pearcey, scale 150 micrometres, source transformations []; copy angle 3.142, phase 3.142 rad, amplitude weight 1; environment free_space. Original effect -0.21088, testable True; failed basic checks: none.

Follow-up routing: C0006H001 -> completed; numerically_reproduced.

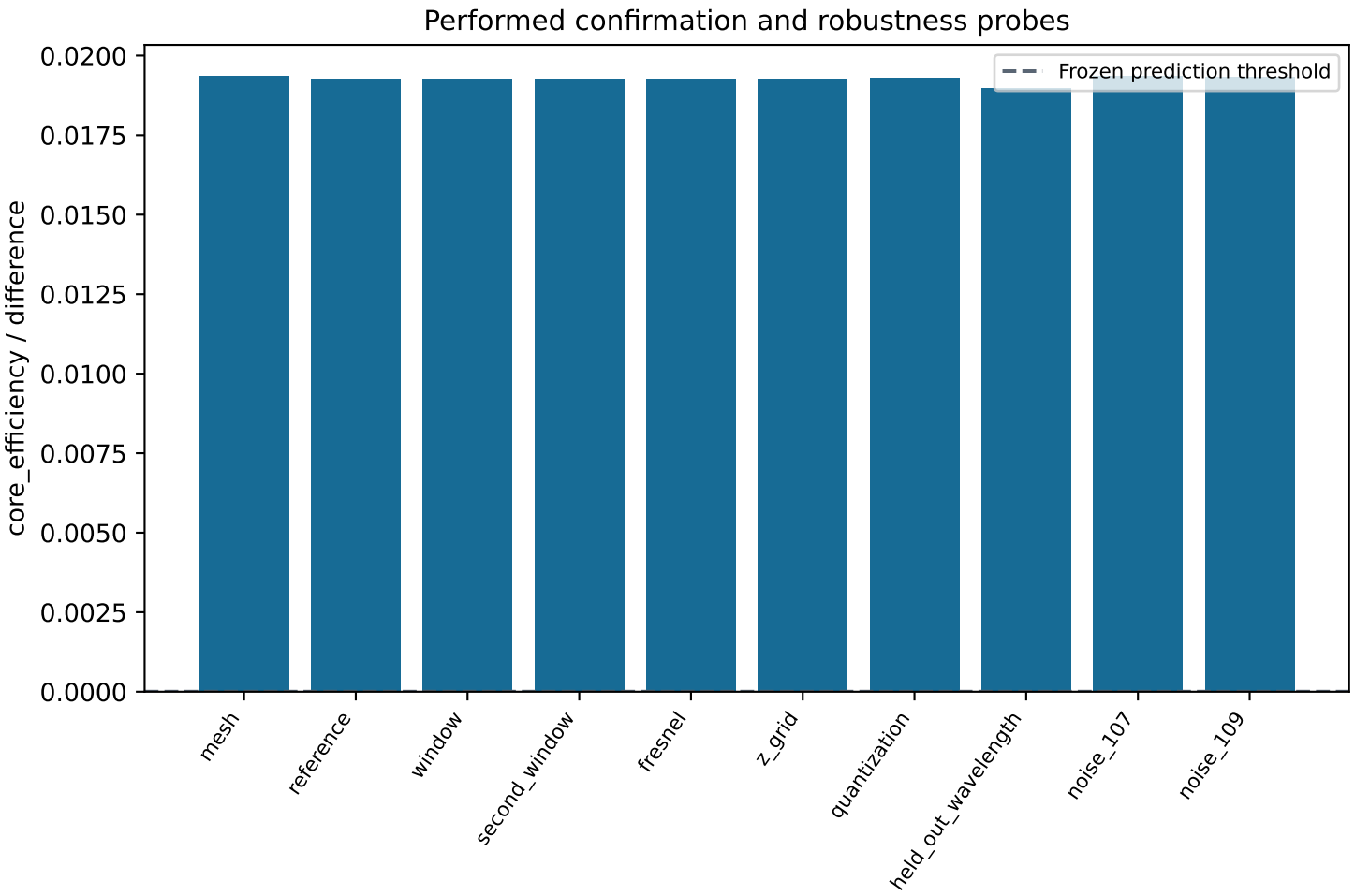
Follow-up routing: C0006H002 -> skipped; unresolved_null_control.

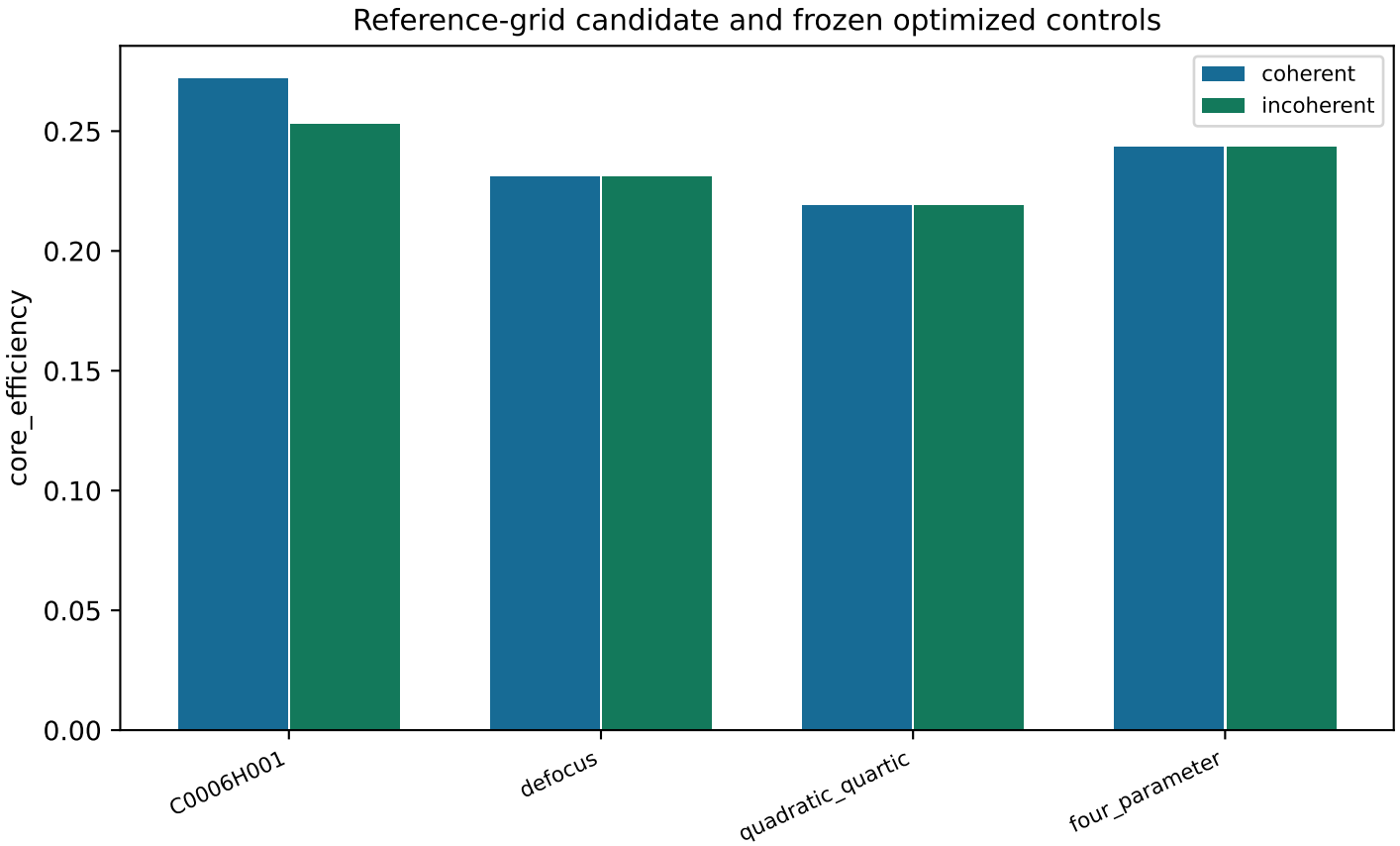
Follow-up routing: C0006H003 -> completed; numerically_reproduced.

C0006H001: numerically_reproduced; reference effect 0.0192613; numerical error floor 0.002. Failed confirmation gates: none. Failed robustness probes: none. Registered direction supported beyond error: True. Frozen optimized controls passed basic checks: True.

C0006H003: numerically_reproduced; reference effect -0.210269; numerical error floor 0.002. Failed confirmation gates: none. Failed robustness probes: none. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: False.

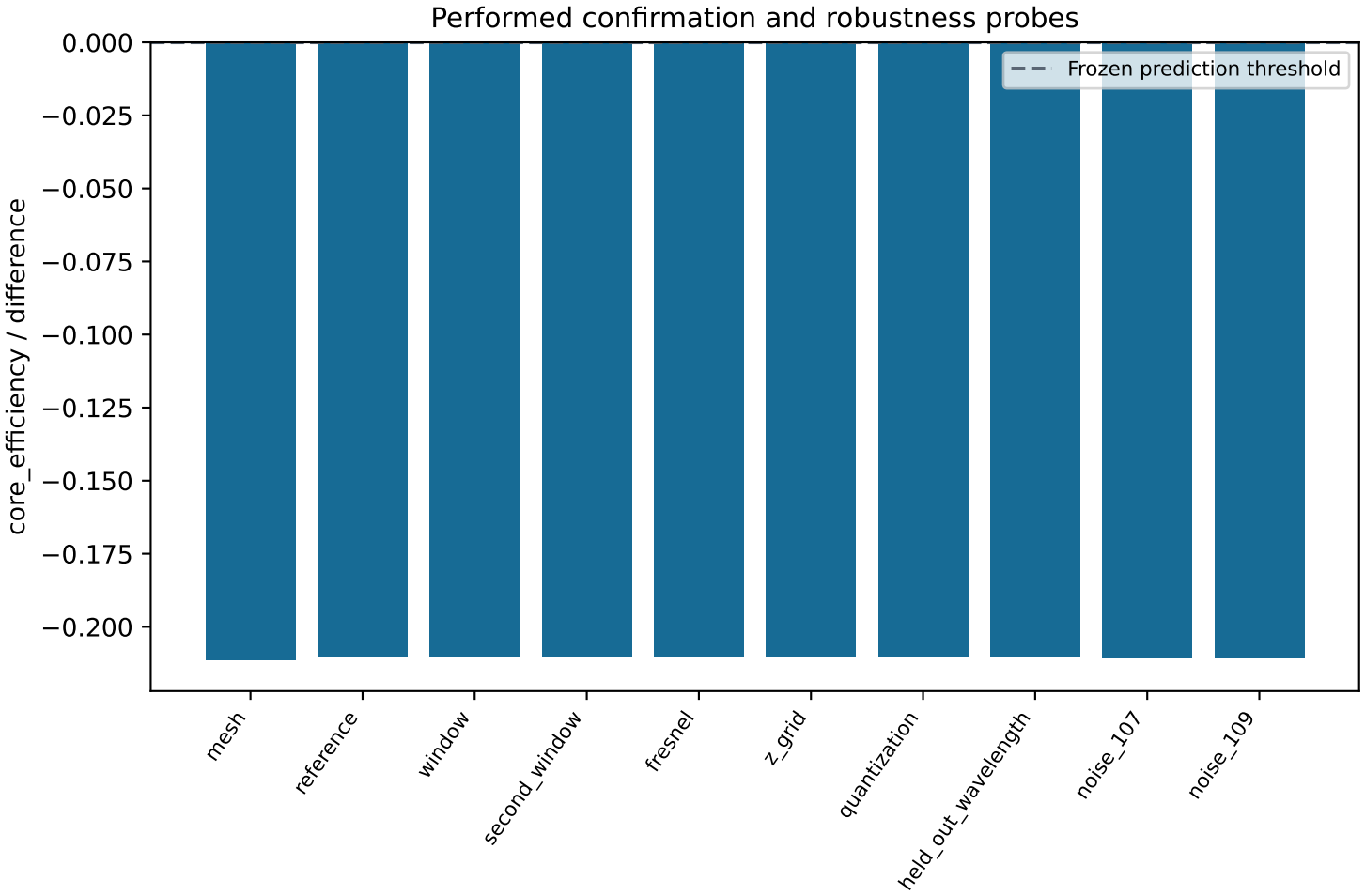
cycle_0006 / C0006H001

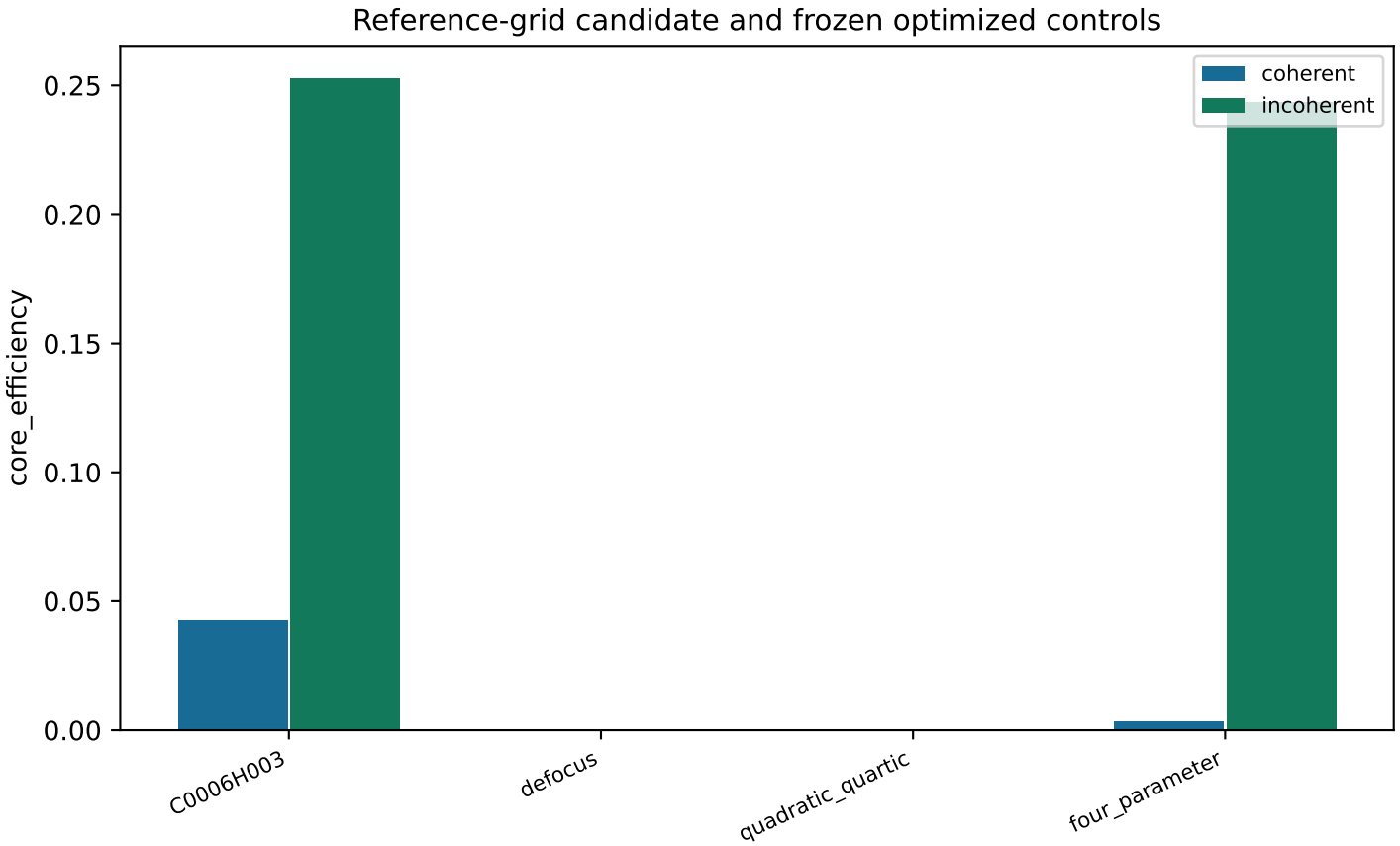




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cycle_0006 / C0006H003





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Evidence batch cycle_0007

Within adequately sampled, finite-aperture finite-energy Airy inputs, does an ideal 60 mm thin lens produce a nonnegative change in peak-trajectory bending relative to matched free-space propagation over 0.02–0.08 m, and is that response confined to particular Airy scale and apodization settings?

Registered proposed mechanism: The lens adds a quadratic phase that changes the transverse phase-space evolution of the Airy field and may alter the curvature of its brightest-pixel trajectory over the frozen propagation window. Aperture truncation, inadequate sampling, and ambiguous or switching brightest-pixel maxima are competing explanations; affected cases must be reported as untestable rather than replaced by another trajectory measure.

Observable: peak_bending_m. Comparison: transformed_vs_base (single versus base). Effect: difference; frozen prediction: >= 0. A comparison across different source cases is not implicit in this interface.

Window: 0.02 to 0.08 m, 11 planes; core radius 110 micrometres. Source artifact: cycle_0007/frozen_protocol.json.

C0007H001: airy, scale 100 micrometres, source transformations []; copy angle 0, phase 0 rad, amplitude weight 1; environment thin_lens. Original effect 0, testable True; failed basic checks: sampling_retention.

C0007H002: airy, scale 150 micrometres, source transformations []; copy angle 0, phase 0 rad, amplitude weight 1; environment thin_lens. Original effect 0, testable True; failed basic checks: sampling_retention.

C0007H003: airy, scale 100 micrometres, source transformations []; copy angle 0, phase 0 rad, amplitude weight 1; environment thin_lens. Original effect 0, testable True; failed basic checks: sampling_retention.

Follow-up routing: C0007H001 -> completed; unconfirmed.

Follow-up routing: C0007H002 -> completed; unconfirmed.

Follow-up routing: C0007H003 -> completed; unconfirmed.

C0007H001: unconfirmed; reference effect 0; numerical error floor 3.125e-06. Failed confirmation gates: mesh_convergence, window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes: quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: True.

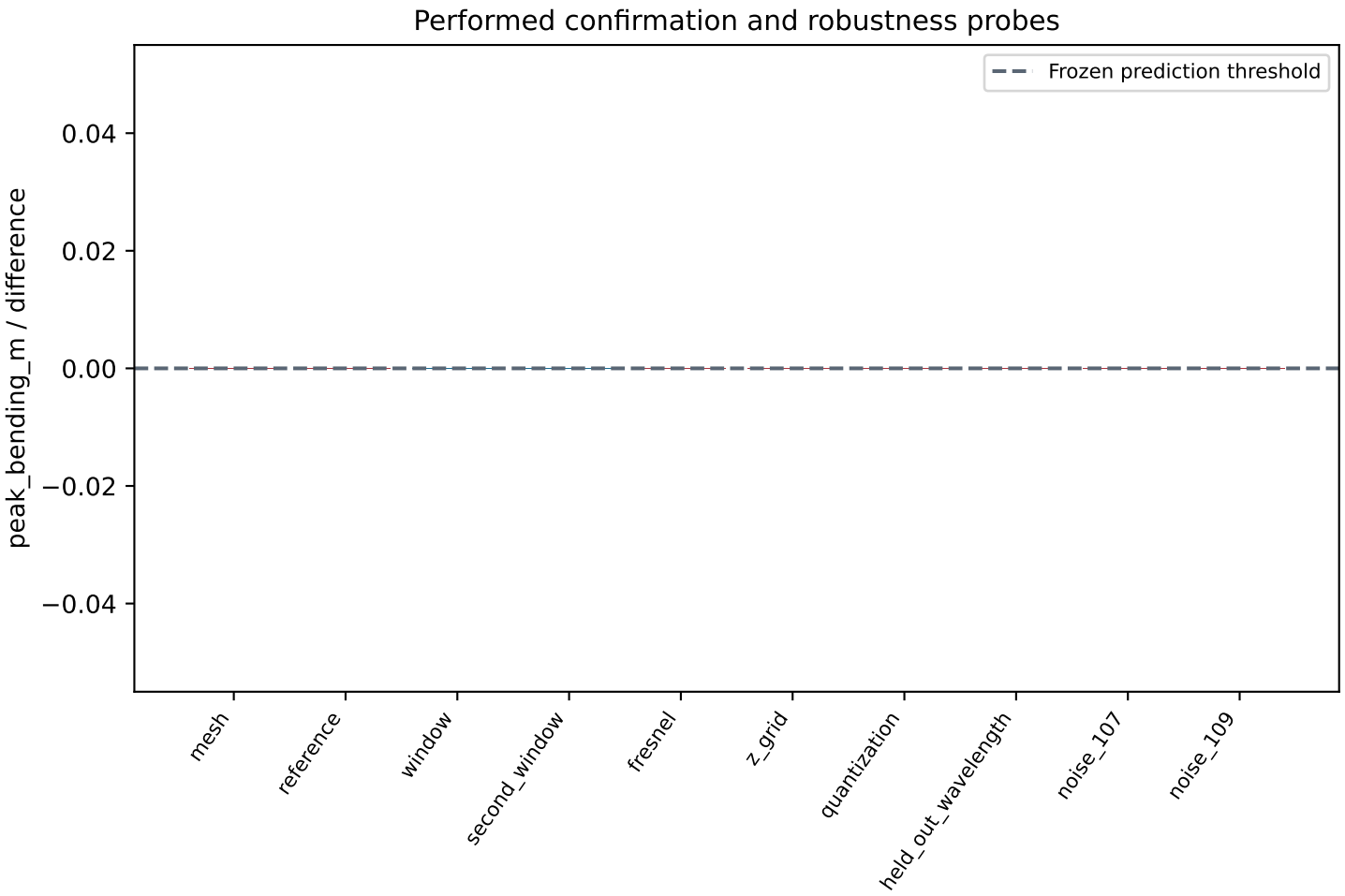
C0007H002: unconfirmed; reference effect 0; numerical error floor 3.125e-06. Failed

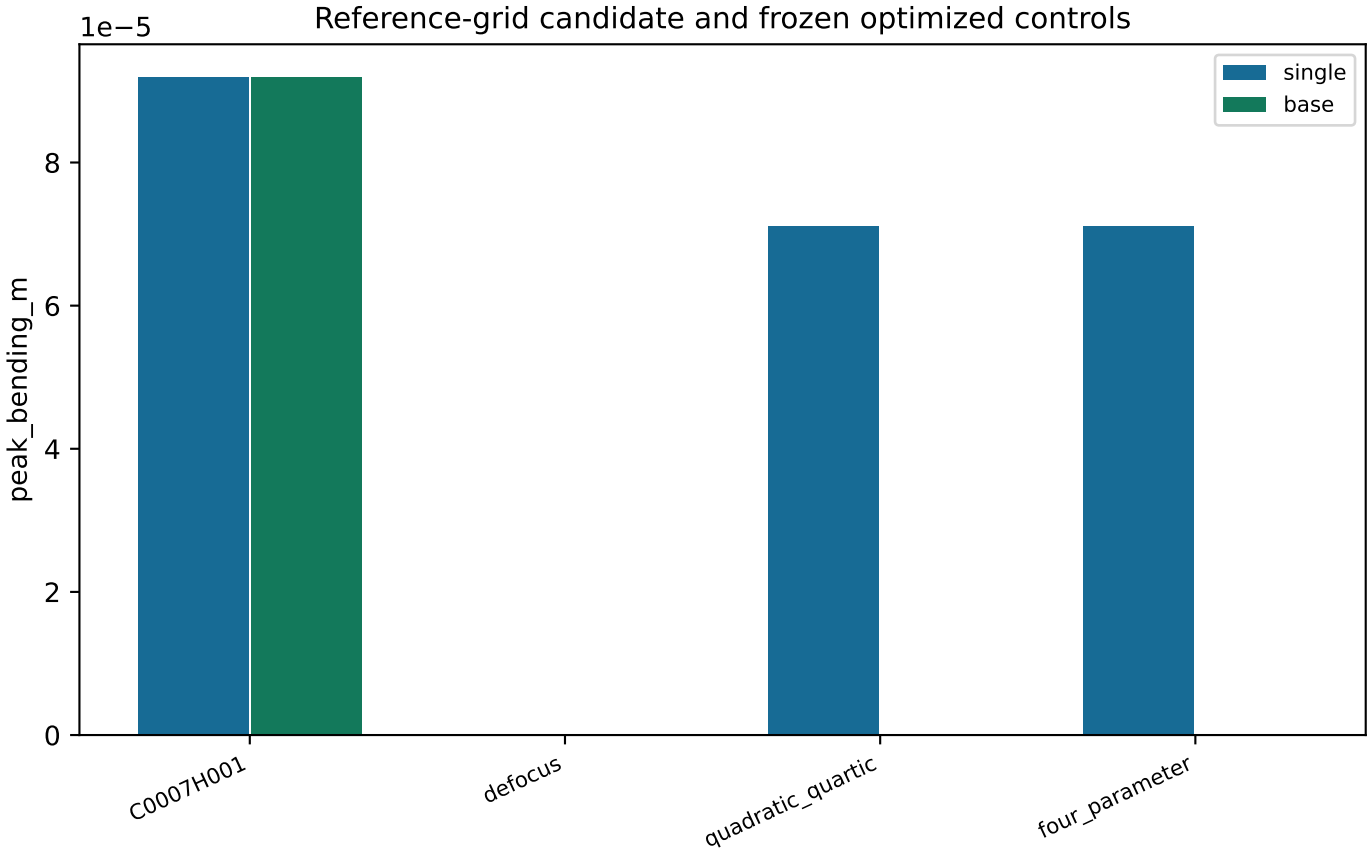
Evidence batch cycle_0007 (continued)

confirmation gates: mesh_convergence, window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes: quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: True.

C0007H003: unconfirmed; reference effect 0; numerical error floor 3.125e-06. Failed confirmation gates: mesh_convergence, window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes: quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: True.

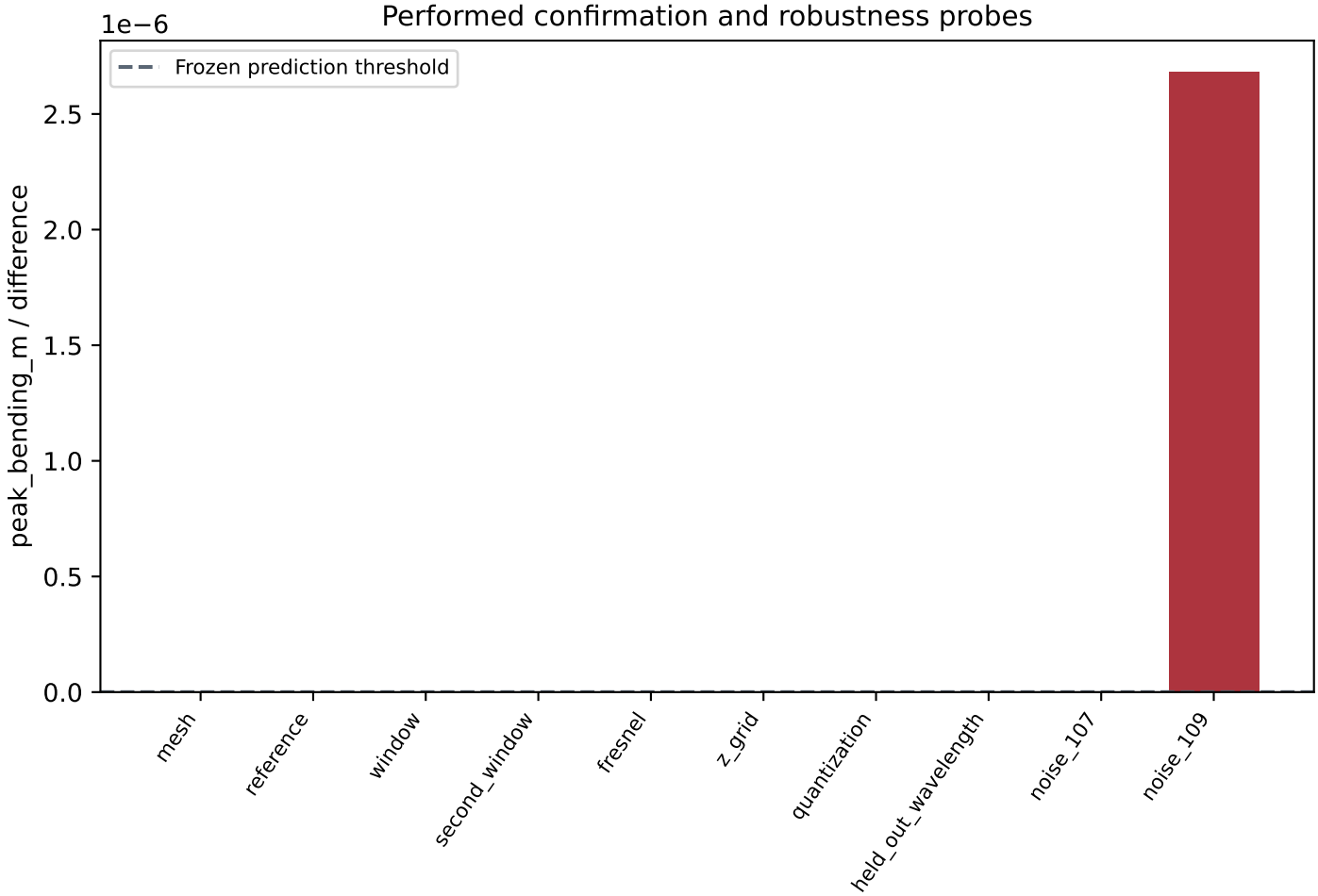
cycle_0007 / C0007H001

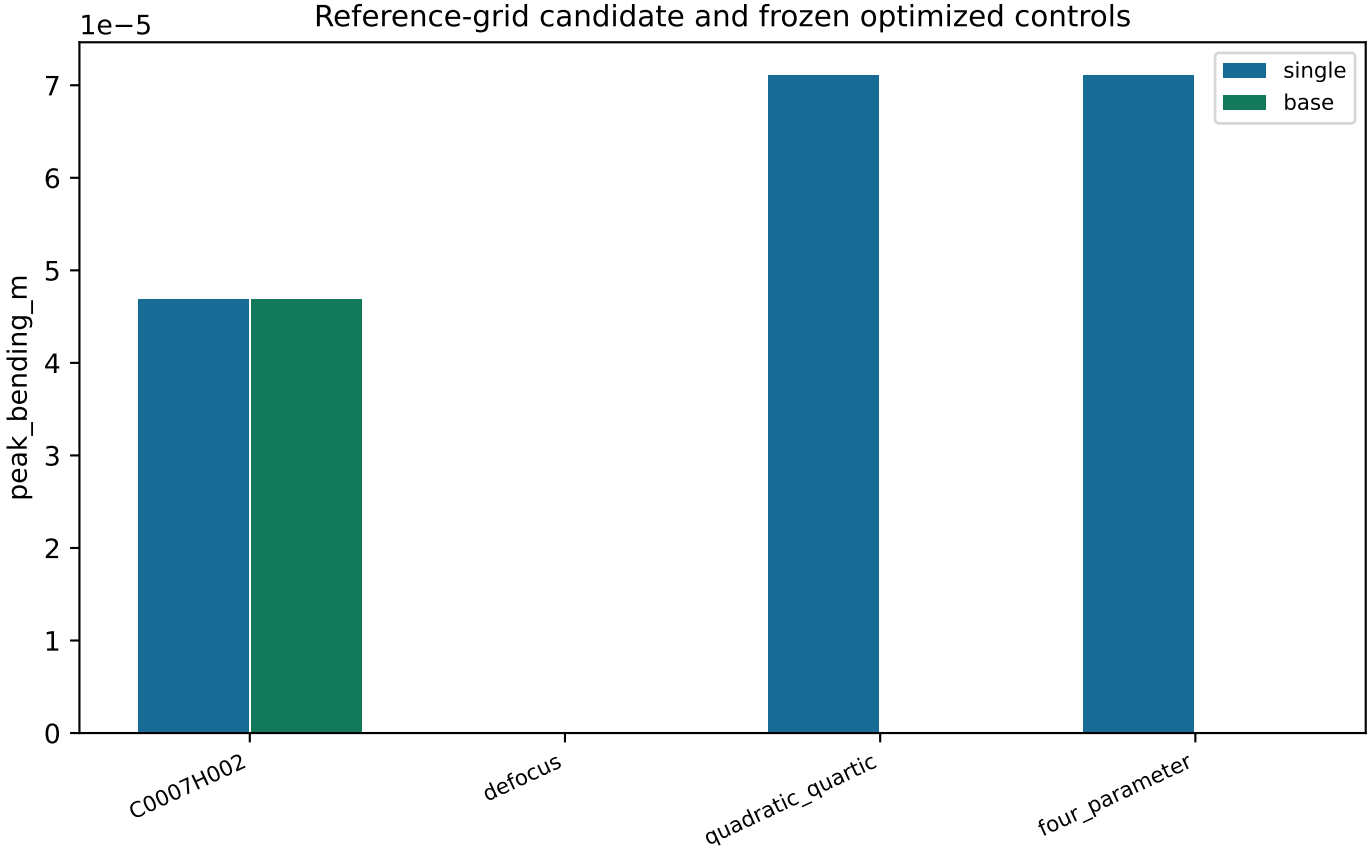




Red bars fail at least one numerical gate. Missing measurements are not zero.
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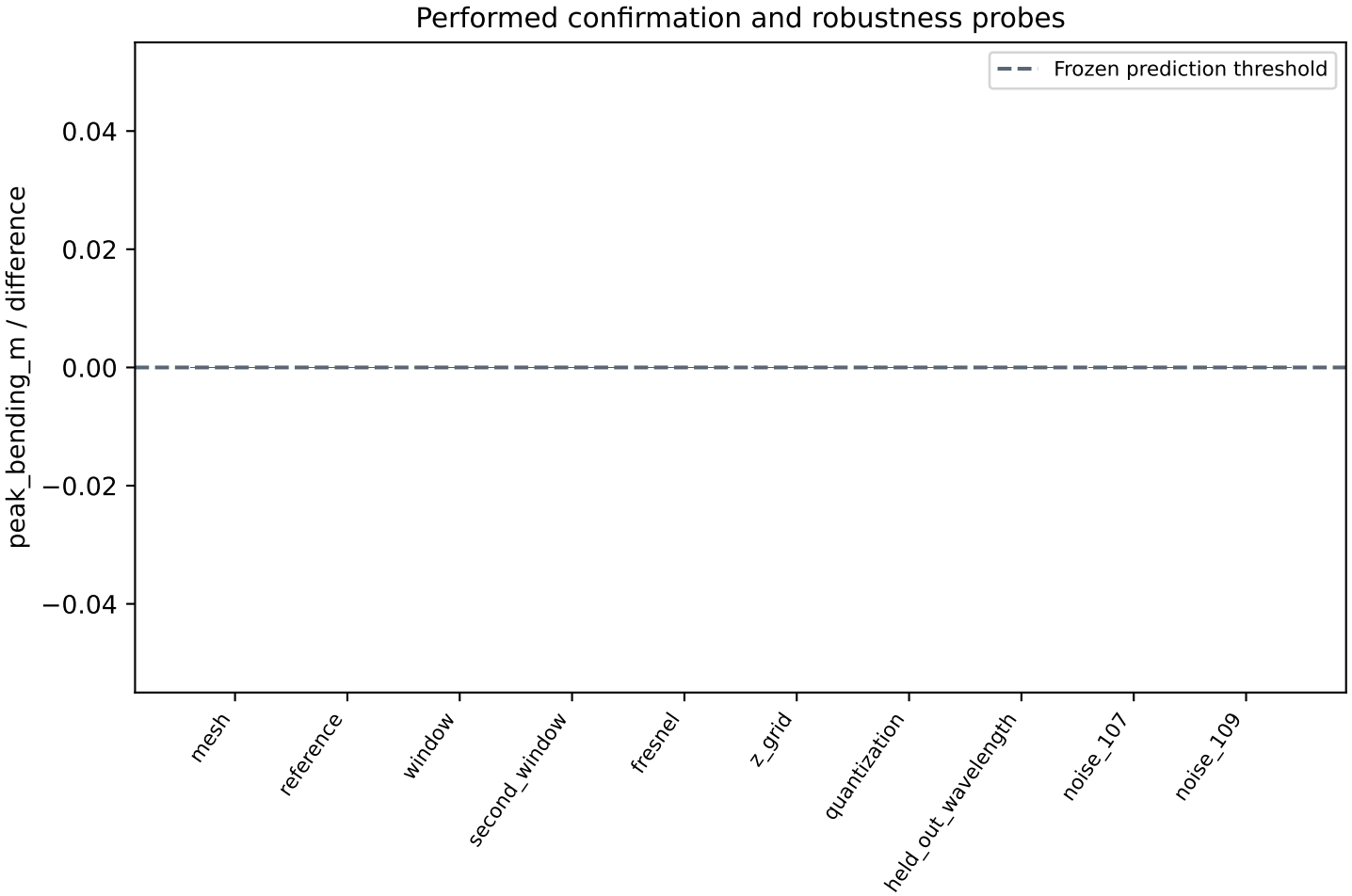
cycle_0007 / C0007H002

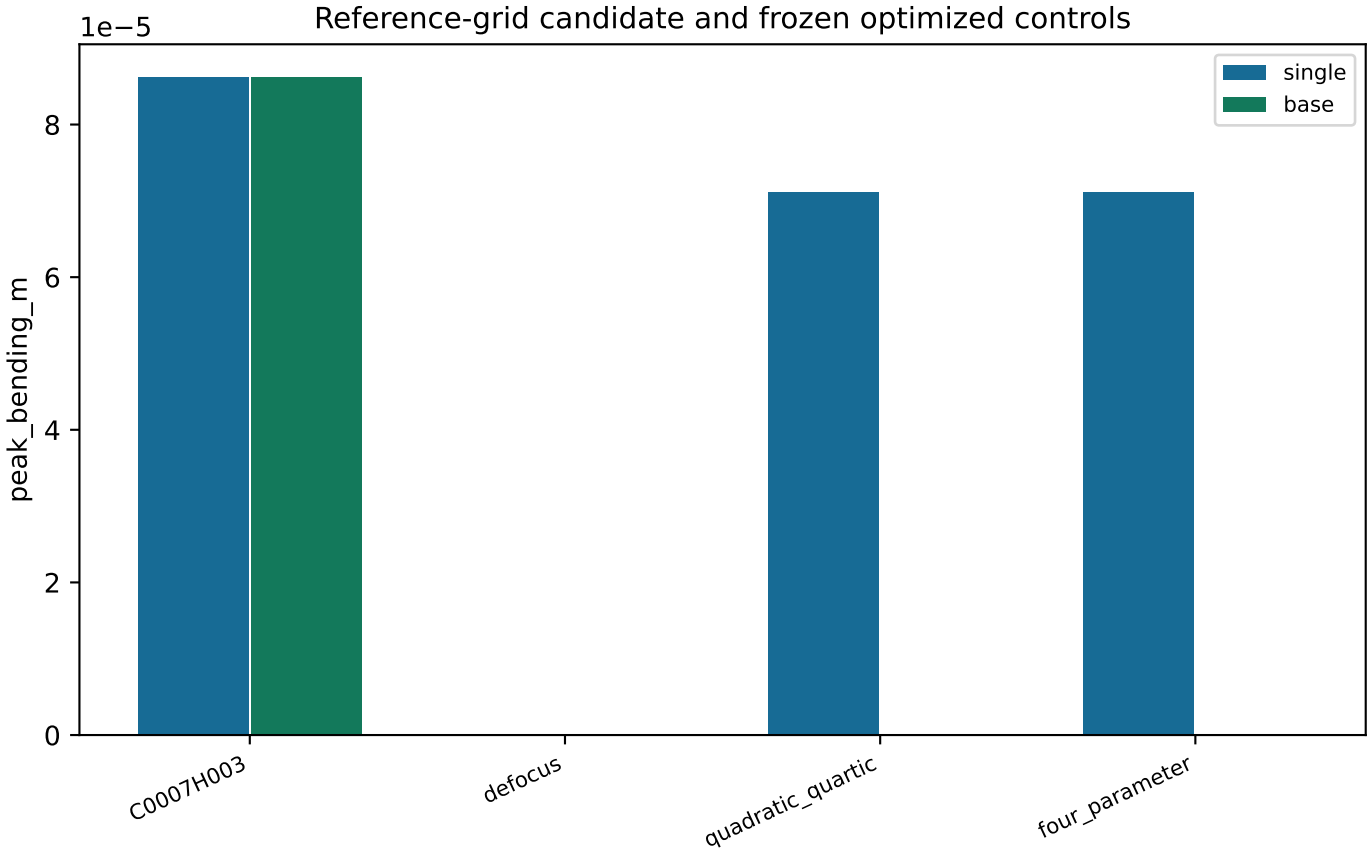




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cycle_0007 / C0007H003





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Controls are re-evaluations of frozen parameters, not exhaustive optima or a novelty test.

Evidence batch cycle_0008

Under equal input power and valid scalar free-space propagation, does coherent superposition of a supported beam with its 180-degree-rotated copy increase median power in a fixed on-axis circular core relative to the single-arm reference, and does the effect depend on beam family or relative phase?

Registered proposed mechanism: Interference between the complex transverse fields of a beam and its rotated copy may redistribute propagated intensity into or out of the fixed on-axis core. Family structure and relative phase may affect the sign and magnitude; numerical validity and equal-power normalization are prerequisites for interpretation.

Observable: core_efficiency. Comparison: coherent_vs_single (coherent versus single). Effect: difference; frozen prediction: >= 0. A comparison across different source cases is not implicit in this interface.

Window: 0.02 to 0.08 m, 11 planes; core radius 100 micrometres. Source artifact: cycle_0008/frozen_protocol.json.

C0008H001: gaussian, scale 100 micrometres, source transformations []; copy angle 3.142, phase 0 rad, amplitude weight 1; environment free_space. Original effect 0, testable True; failed basic checks: none.

C0008H002: airy, scale 150 micrometres, source transformations []; copy angle 3.142, phase 0 rad, amplitude weight 1; environment free_space. Original effect 0.0367003, testable True; failed basic checks: window_edges.

C0008H003: pearcey, scale 200 micrometres, source transformations []; copy angle 3.142, phase 3.142 rad, amplitude weight 1; environment free_space. Original effect -0.193119, testable True; failed basic checks: none.

Follow-up routing: C0008H001 -> skipped; known_identity.

Follow-up routing: C0008H002 -> completed; unconfirmed.

Follow-up routing: C0008H003 -> completed; numerically_reproduced.

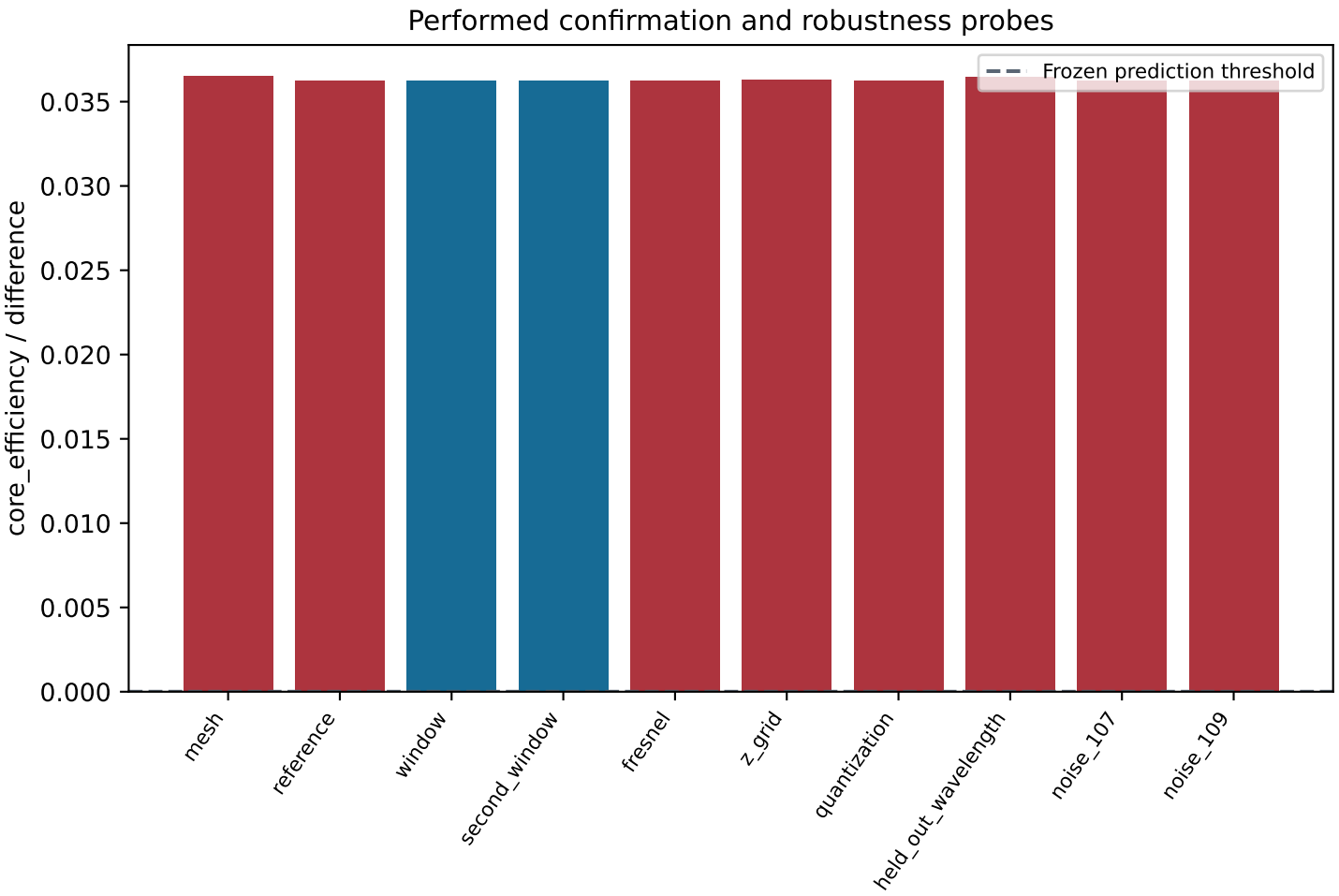
C0008H002: unconfirmed; reference effect 0.0362368; numerical error floor 0.002. Failed confirmation gates: mesh_convergence, window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes: quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: True.

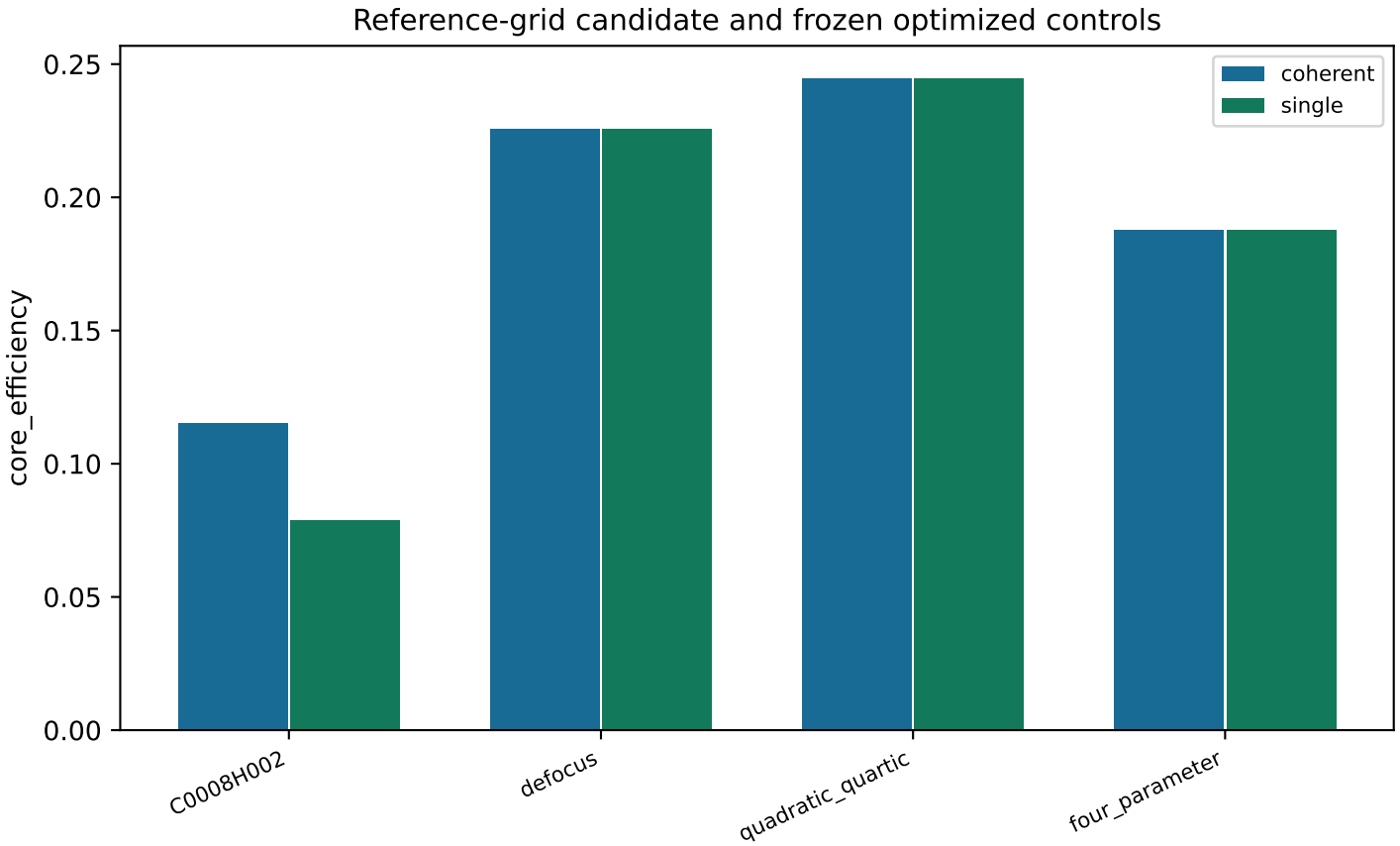
C0008H003: numerically_reproduced; reference effect -0.192446; numerical error floor 0.002. Failed confirmation gates: none. Failed robustness probes: none. Registered direction

Evidence batch cycle_0008 (continued)

supported beyond error: False. Frozen optimized controls passed basic checks: False.

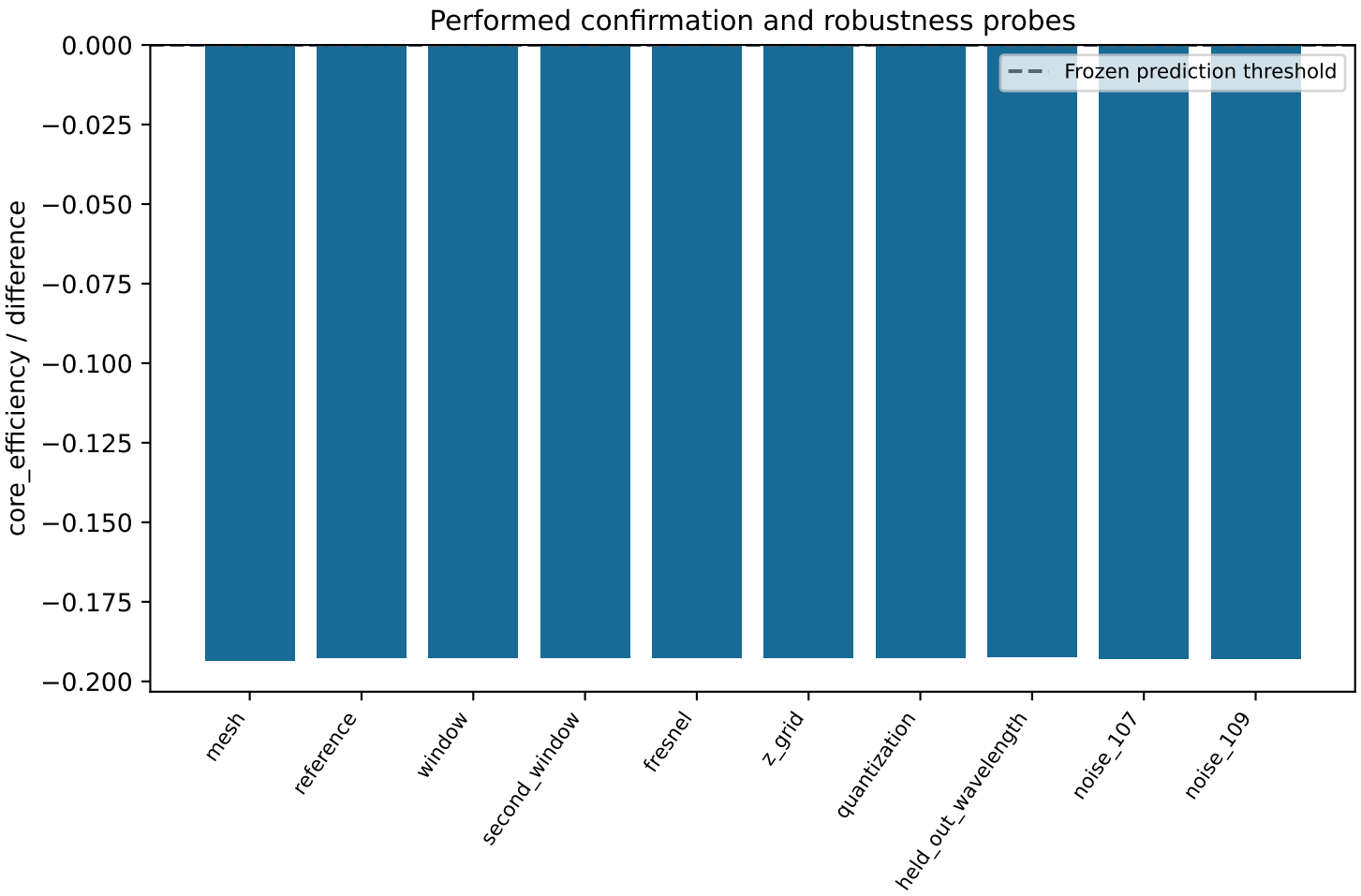
cycle_0008 / C0008H002

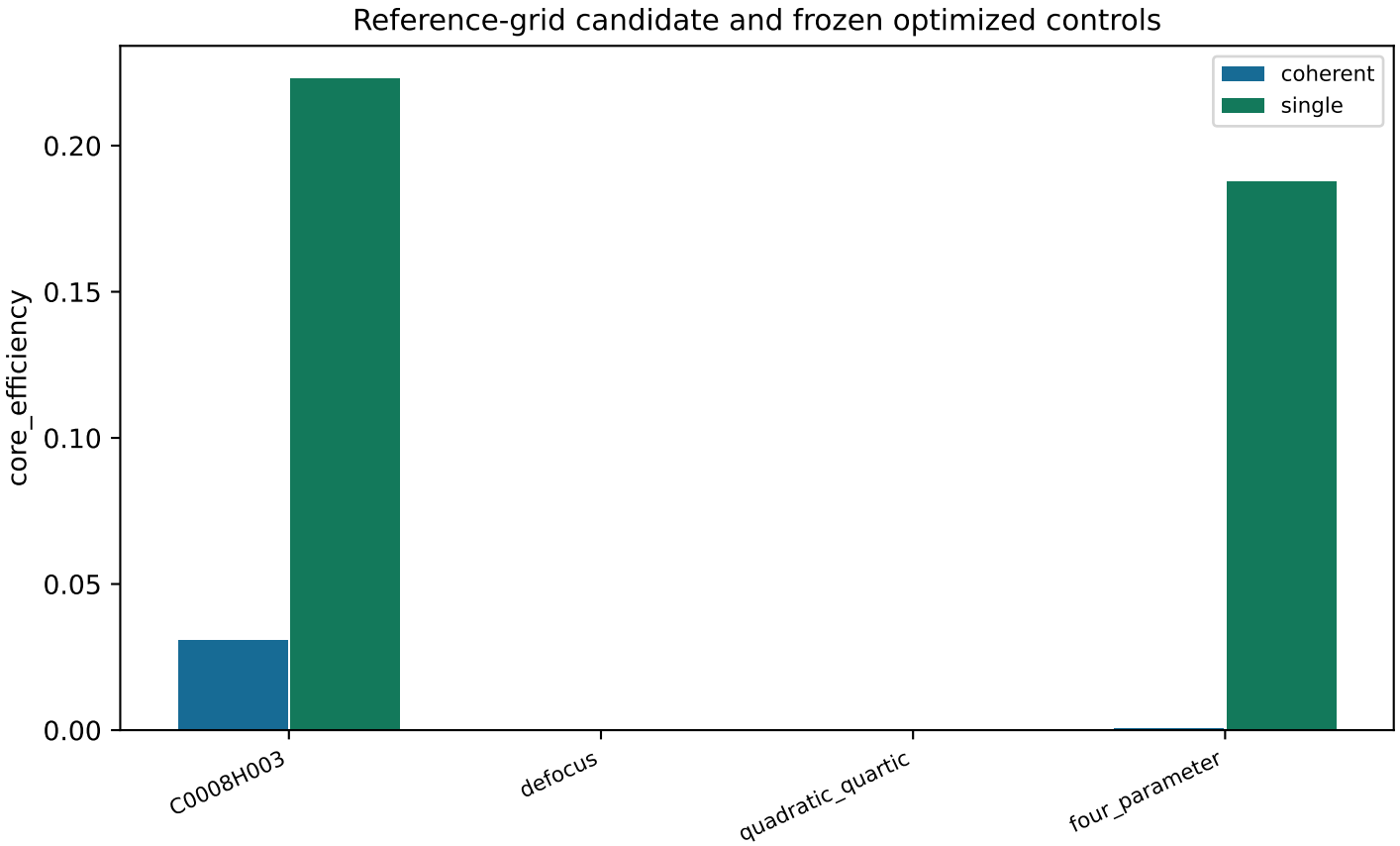




Red bars fail at least one numerical gate. Missing measurements are not zero.
Controls are re-evaluations of frozen parameters, not exhaustive optima or a novelty test.

cycle_0008 / C0008H003





Red bars fail at least one numerical gate. Missing measurements are not zero.
Controls are re-evaluations of frozen parameters, not exhaustive optima or a novelty test.

Evidence batch cycle_0009

For a fixed apodized Pearcey field coherently composed with its 180-degree-rotated copy, does integer vortex charge change the relative-phase dependence of median fixed-core efficiency, beyond changes attributable to the transformed single arm or constituent identity?

Registered proposed mechanism: Applying an integer vortex phase changes the Pearcey field's transverse phase distribution; coherent addition with a rotated copy can therefore alter interference within the fixed on-axis core as relative phase varies. The registered comparisons test core efficiency against a matched uncomposed single arm. They are directional tests, not by themselves a measurement of the charge-by-phase interaction or evidence of topological protection.

Observable: core_efficiency. Comparison: coherent_vs_single (coherent versus single). Effect: difference; frozen prediction: <= 0. A comparison across different source cases is not implicit in this interface.

Window: 0.04 to 0.08 m, 11 planes; core radius 100 micrometres. Source artifact: cycle_0009/frozen_protocol.json.

C0009H001: pearcey, scale 200 micrometres, source transformations []; copy angle 3.142, phase 3.142 rad, amplitude weight 1; environment free_space. Original effect -0.193648, testable True; failed basic checks: none.

C0009H002: pearcey, scale 200 micrometres, source transformations [{'kind': 'vortex', 'amount': 1}]; copy angle 3.142, phase 1.571 rad, amplitude weight 1; environment free_space. Original effect -1.38778e-17, testable True; failed basic checks: none.

C0009H003: pearcey, scale 200 micrometres, source transformations [{'kind': 'vortex', 'amount': 1}]; copy angle 3.142, phase 3.142 rad, amplitude weight 1; environment free_space. Original effect 0.00219853, testable True; failed basic checks: none.

Follow-up routing: C0009H001 -> completed; numerically_reproduced.

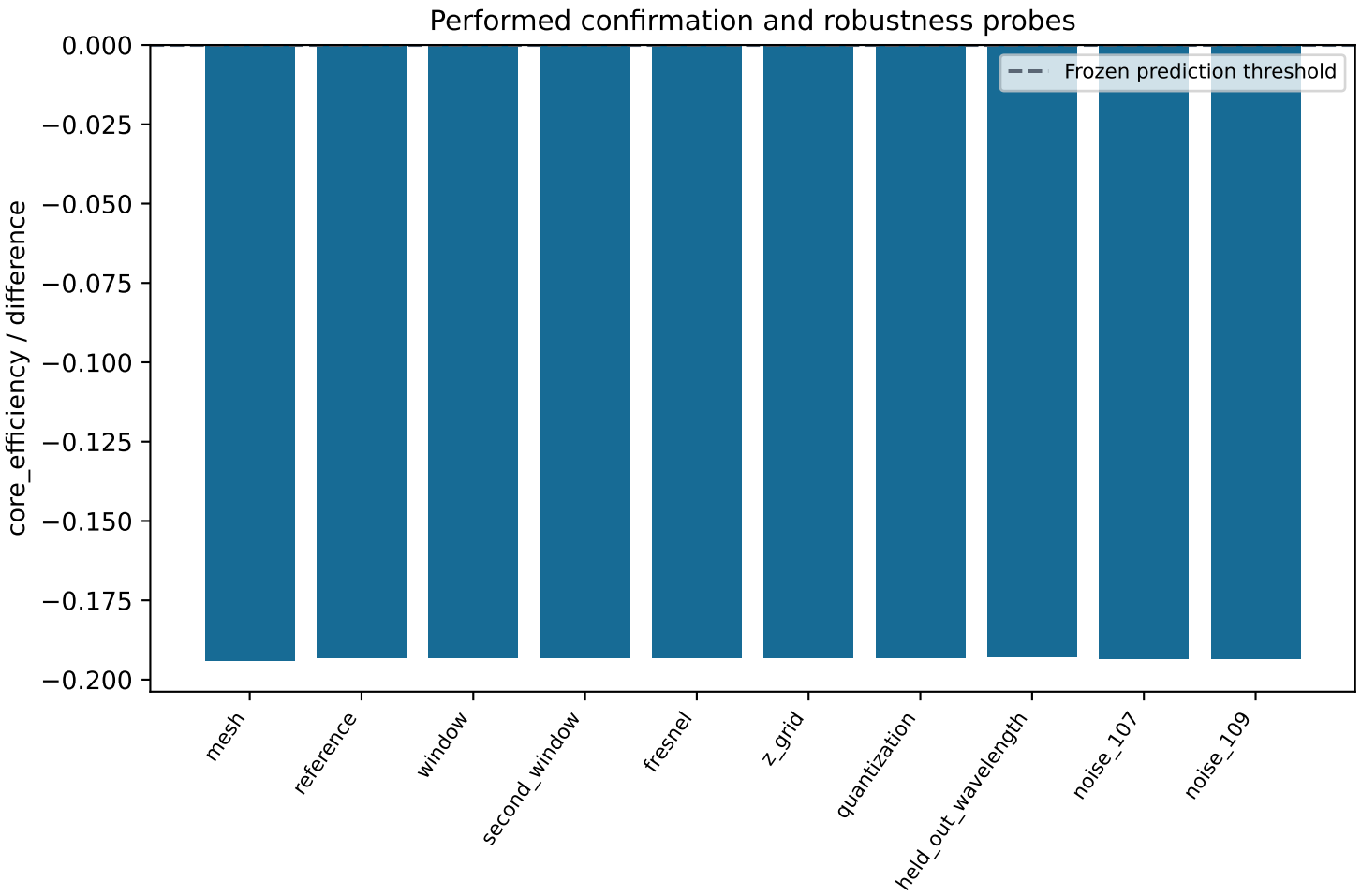
Follow-up routing: C0009H002 -> skipped; unresolved_null_control.

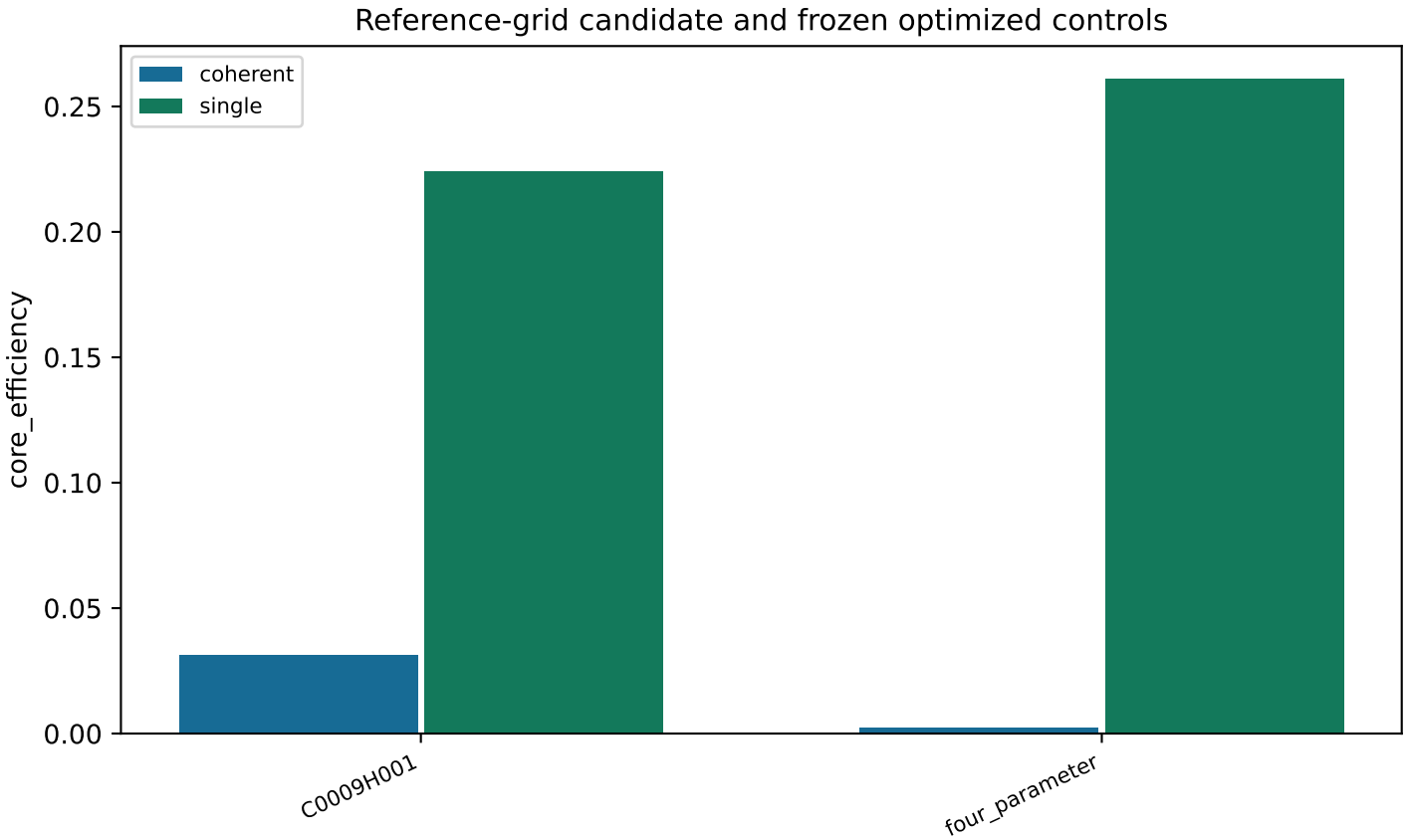
Follow-up routing: C0009H003 -> completed; numerically_reproduced.

C0009H001: numerically_reproduced; reference effect -0.192977; numerical error floor 0.002. Failed confirmation gates: none. Failed robustness probes: none. Registered direction supported beyond error: True. Frozen optimized controls passed basic checks: True.

C0009H003: numerically_reproduced; reference effect 0.00219489; numerical error floor 0.002. Failed confirmation gates: none. Failed robustness probes: none. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: True.

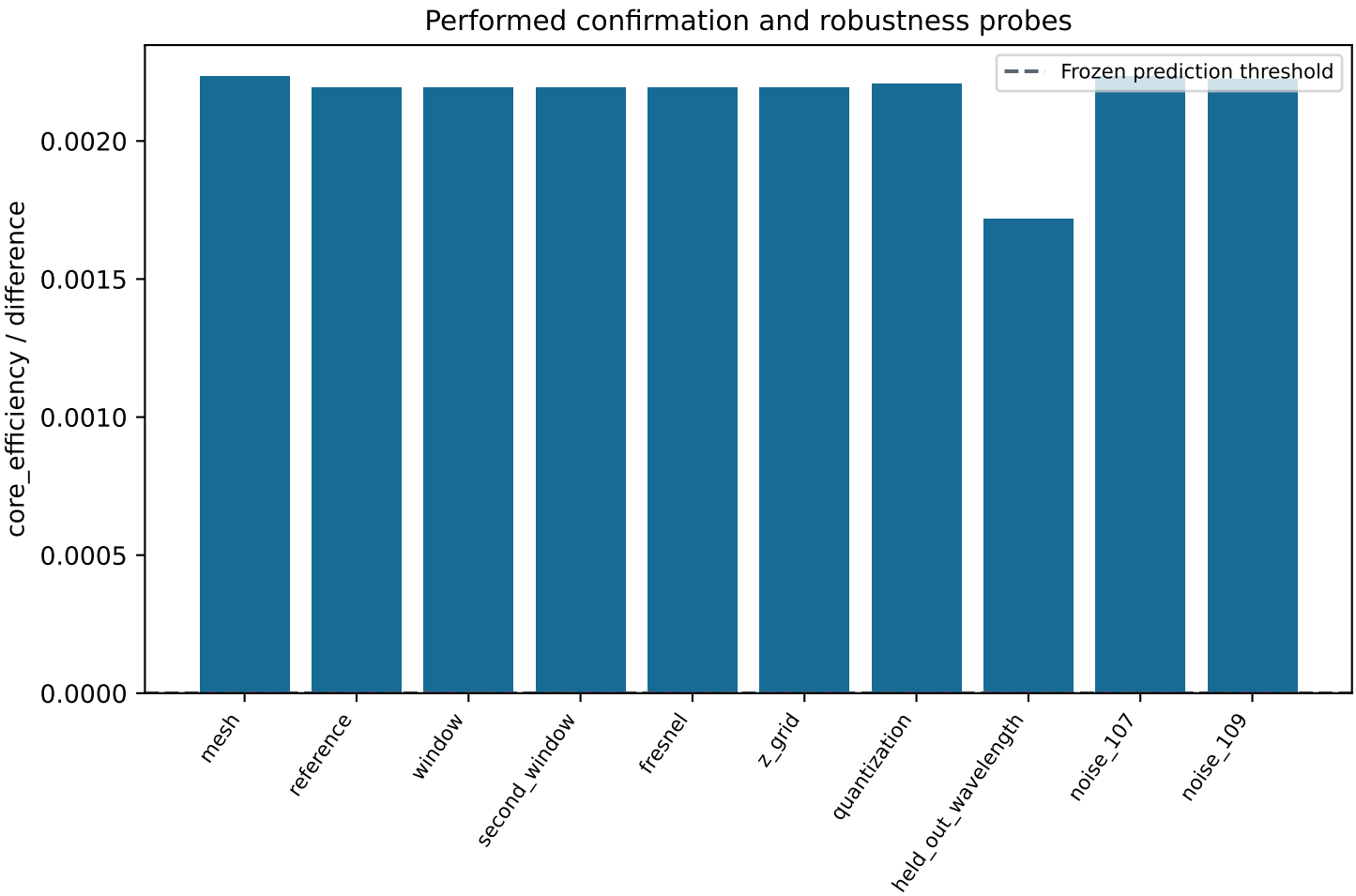
cycle_0009 / C0009H001

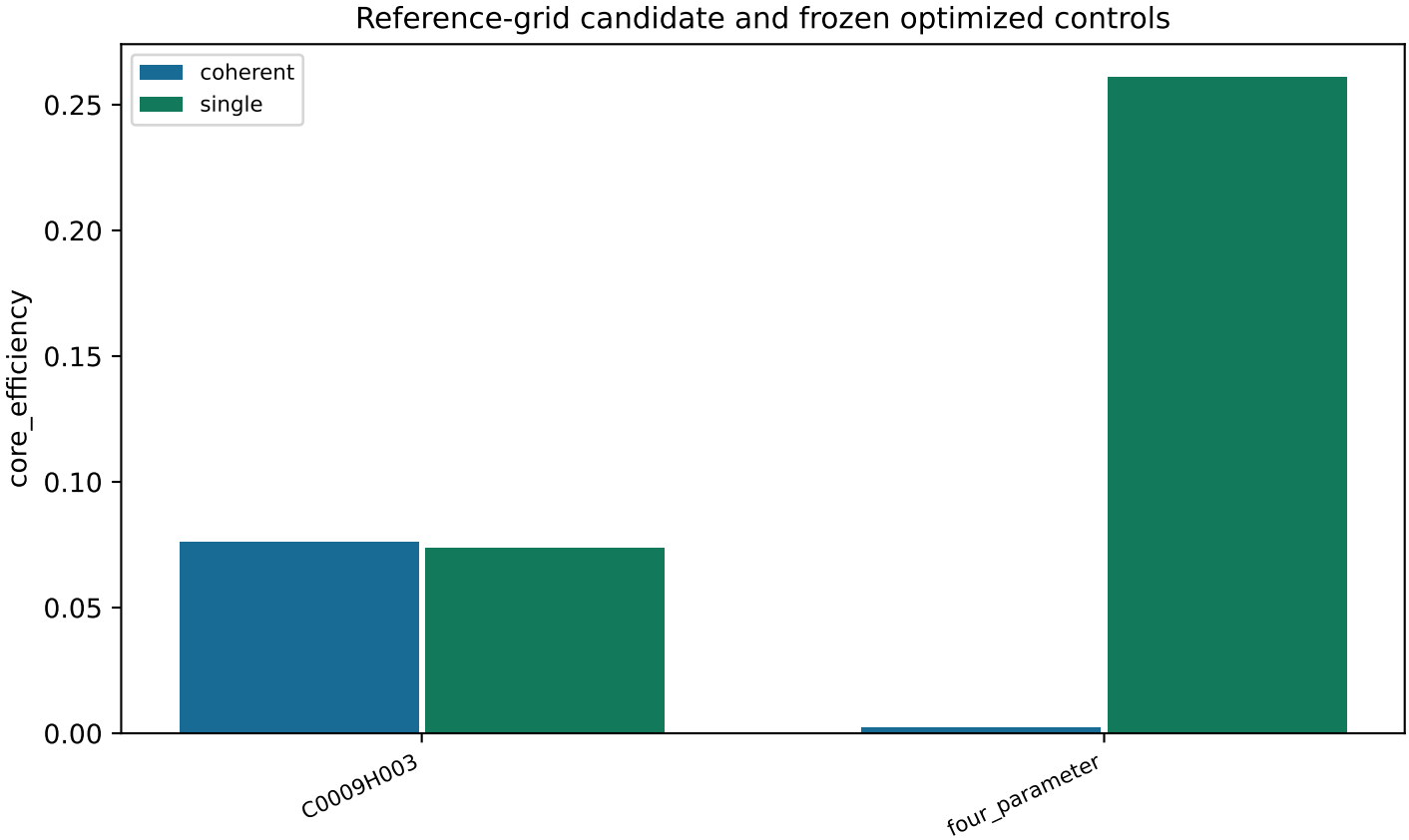




Red bars fail at least one numerical gate. Missing measurements are not zero.
Controls are re-evaluations of frozen parameters, not exhaustive optima or a novelty test.

cycle_0009 / C0009H003





Red bars fail at least one numerical gate. Missing measurements are not zero.
Controls are re-evaluations of frozen parameters, not exhaustive optima or a novelty test.

Evidence batch cycle_0011

For a p=0, charge-l=1 vortex field, does coherent addition with an equal-amplitude 180-degree rotated copy produce peak_intensity no greater than the single-arm reference over free-space propagation from 0.03 to 0.09 m, as predicted by odd rotational parity?

Registered proposed mechanism: A 180-degree rotation multiplies a vortex field of integer charge l by (-1)^l. For odd charge, the rotated field has opposite sign, so an equal-amplitude, zero-relative-phase coherent sum predicts destructive cancellation. The three cases test this prediction for odd-charge vortex inputs; even-charge and Gaussian parity controls are not encoded as directional hypotheses in this protocol.

Observable: peak_intensity. Comparison: coherent_vs_single (coherent versus single). Effect: ratio; frozen prediction: <= 1.0. A comparison across different source cases is not implicit in this interface.

Window: 0.03 to 0.09 m, 11 planes; core radius 110 micrometres. Source artifact: cycle_0011/frozen_protocol.json.

C0011H001: vortex, scale 100 micrometres, source transformations []; copy angle 3.142, phase 0 rad, amplitude weight 1; environment free_space. Original effect not defined, testable False; failed basic checks: none.

C0011H002: vortex, scale 100 micrometres, source transformations []; copy angle 3.142, phase 0 rad, amplitude weight 1; environment free_space. Original effect not defined, testable False; failed basic checks: none.

C0011H003: vortex, scale 100 micrometres, source transformations []; copy angle 3.142, phase 0 rad, amplitude weight 1; environment free_space. Original effect not defined, testable False; failed basic checks: none.

Follow-up routing: C0011H001 -> skipped; execution_error_or_zero_power_comparison.

Follow-up routing: C0011H002 -> skipped; execution_error_or_zero_power_comparison.

Follow-up routing: C0011H003 -> skipped; execution_error_or_zero_power_comparison.

Evidence batch cycle_0013

For supported structured inputs that produce two prominent ridges in the registered transverse profile, does coherent superposition with a 180-degree rotated copy increase trusted valley_filling relative to the corresponding single arm in a fixed, numerically valid free-space observation window?

Registered proposed mechanism: The rotated copy can overlap the original with a spatially varying relative phase. Coherent interference may add intensity in inter-ridge valleys, while incoherent addition removes cross terms. Whether the overlap phase and ridge geometry raise valley filling is conditional and must be tested.

Observable: valley_filling. Comparison: coherent_vs_single (coherent versus single). Effect: ratio; frozen prediction: >= 1.0. A comparison across different source cases is not implicit in this interface.

Window: 0.04 to 0.1 m, 11 planes; core radius 110 micrometres. Source artifact: cycle_0013/frozen_protocol.json.

C0013H001: pearcey, scale 200 micrometres, source transformations []; copy angle 3.142, phase 0 rad, amplitude weight 1; environment free_space. Original effect not defined, testable False; failed basic checks: none.

C0013H002: butterfly, scale 200 micrometres, source transformations []; copy angle 3.142, phase 1.571 rad, amplitude weight 1; environment free_space. Original effect not defined, testable False; failed basic checks: none.

C0013H003: airy, scale 100 micrometres, source transformations []; copy angle 3.142, phase 3.142 rad, amplitude weight 1; environment free_space. Original effect 5.63232, testable True; failed basic checks: window_edges.

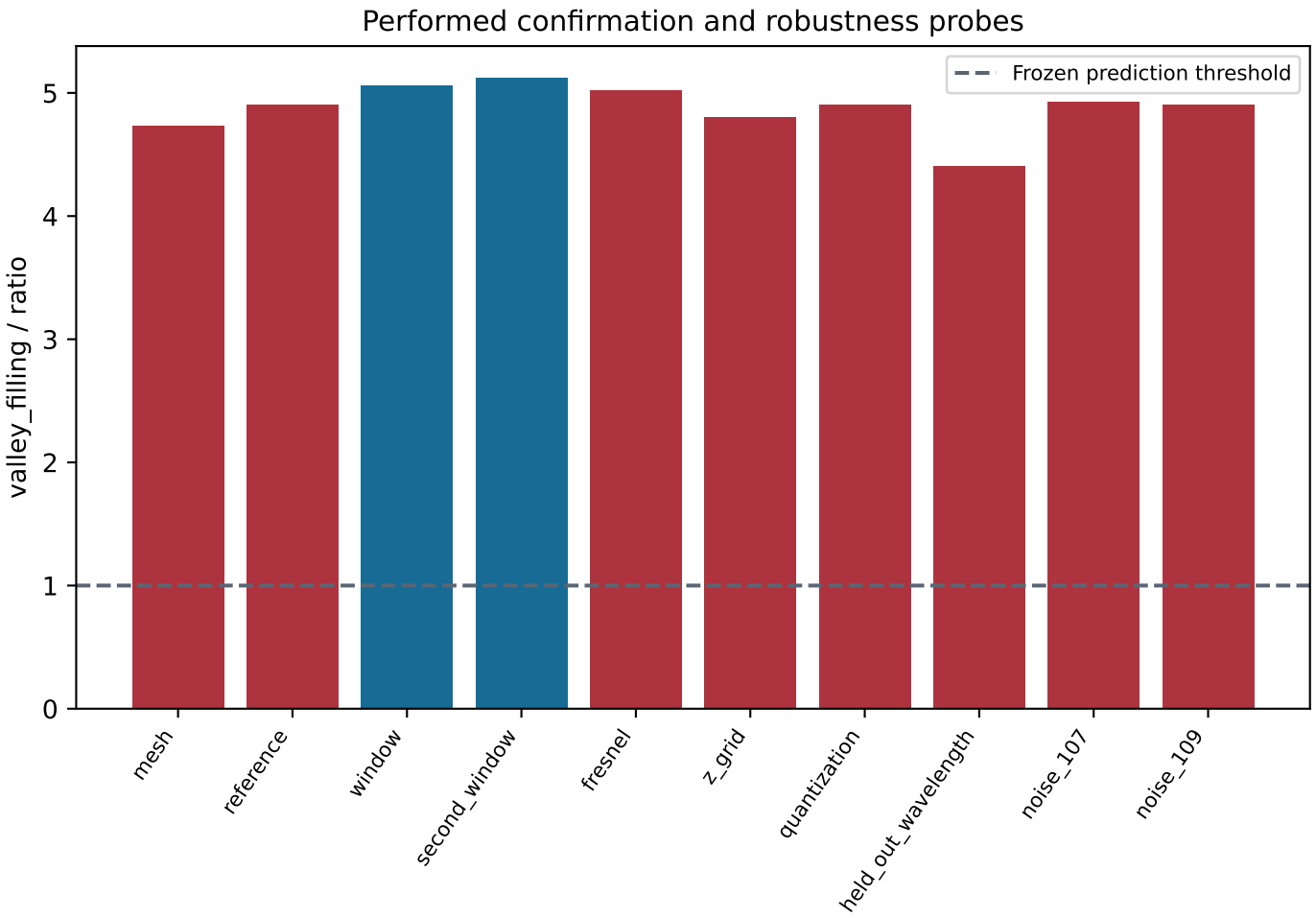
Follow-up routing: C0013H001 -> skipped; untestable_observable.

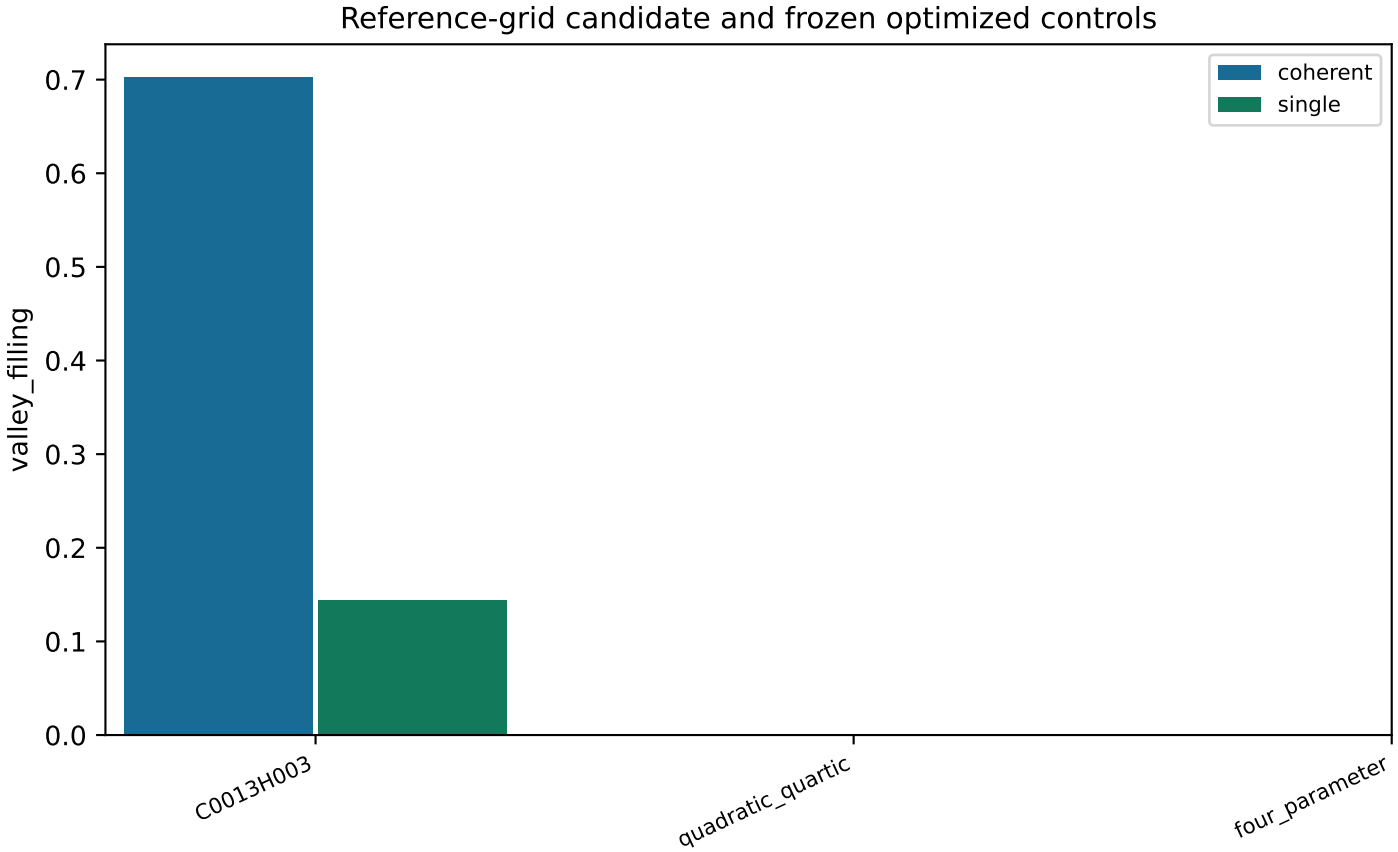
Follow-up routing: C0013H002 -> skipped; untestable_observable.

Follow-up routing: C0013H003 -> completed; unconfirmed.

C0013H003: unconfirmed; reference effect 4.90154; numerical error floor 0.222518. Failed confirmation gates: mesh_convergence, window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes: quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: False.

cycle_0013 / C0013H003





Red bars fail at least one numerical gate. Missing measurements are not zero.
Controls are re-evaluations of frozen parameters, not exhaustive optima or a novelty test.

Evidence batch cycle_0014

In numerically converged finite-energy Airy-beam simulations, does relative phase in a coherent 180-degree rotated-copy composition systematically change intensity_overlap with the single-arm reference, and can that change be distinguished from the matched incoherent intensity sum?

Registered proposed mechanism: Relative phase changes the interference cross term between the Airy field and its 180-degree rotated copy. Their propagated spatial overlap may redistribute transverse intensity and alter its cosine similarity to the single-arm intensity pattern. The matched incoherent comparison tests whether any change depends on coherent interference.

Observable: intensity_overlap. Comparison: coherent_vs_single (coherent versus single). Effect: ratio; frozen prediction: >= 1.0. A comparison across different source cases is not implicit in this interface.

Window: 0.03 to 0.07 m, 11 planes; core radius 100 micrometres. Source artifact: cycle_0014/frozen_protocol.json.

C0014H001: airy, scale 100 micrometres, source transformations []; copy angle 3.142, phase 0 rad, amplitude weight 1; environment free_space. Original effect 0.737028, testable True; failed basic checks: window_edges.

C0014H002: airy, scale 100 micrometres, source transformations []; copy angle 3.142, phase 1.571 rad, amplitude weight 1; environment free_space. Original effect 0.64643, testable True; failed basic checks: window_edges.

C0014H003: airy, scale 100 micrometres, source transformations []; copy angle 3.142, phase 3.142 rad, amplitude weight 1; environment free_space. Original effect 0.682967, testable True; failed basic checks: window_edges.

Follow-up routing: C0014H001 -> completed; unconfirmed.

Follow-up routing: C0014H002 -> completed; unconfirmed.

Follow-up routing: C0014H003 -> completed; unconfirmed.

C0014H001: unconfirmed; reference effect 0.737047; numerical error floor 0.01. Failed confirmation gates: mesh_convergence, window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes: quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: True.

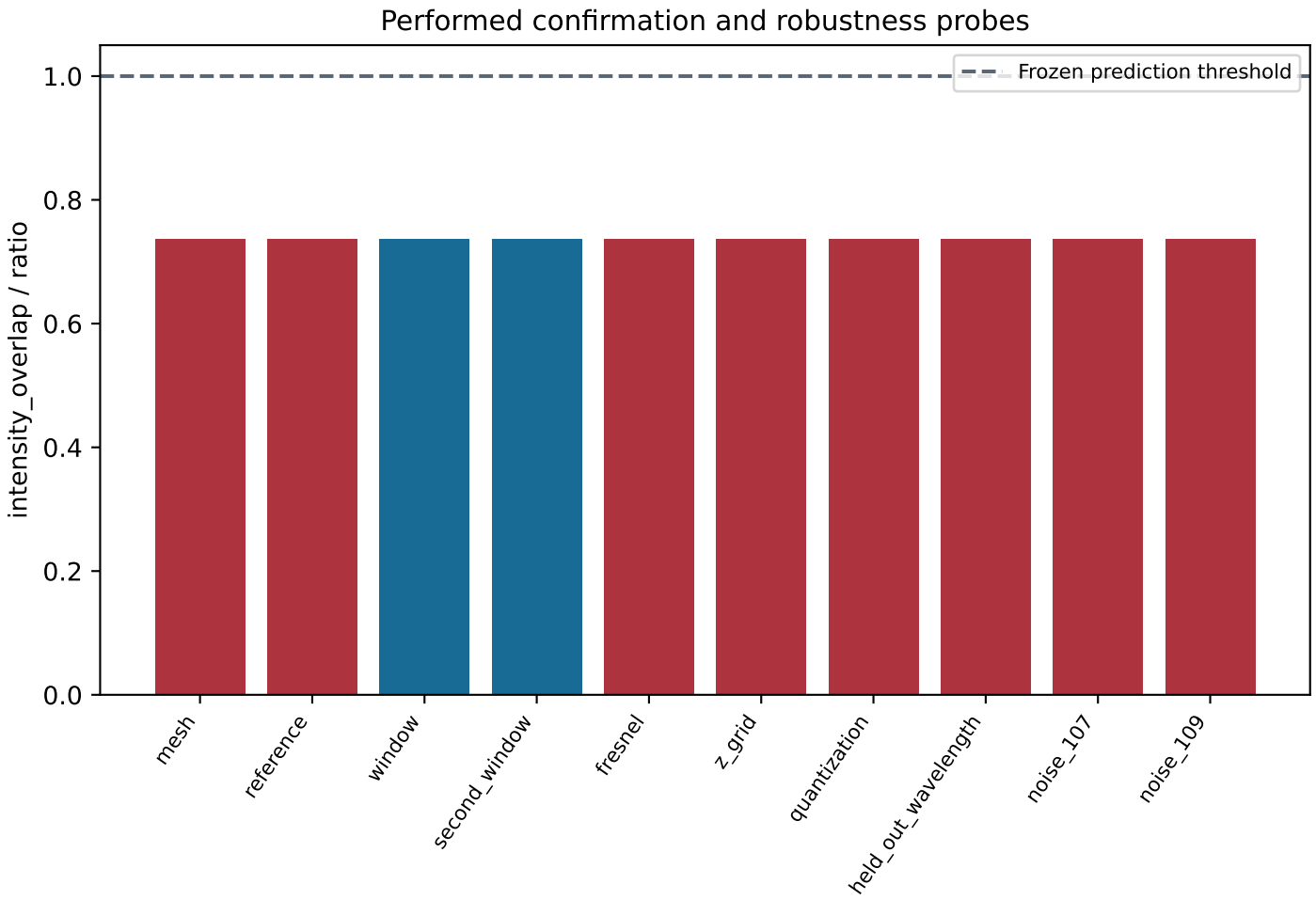
C0014H002: unconfirmed; reference effect 0.64644; numerical error floor 0.01. Failed confirmation gates: mesh_convergence, window_convergence, z_convergence,

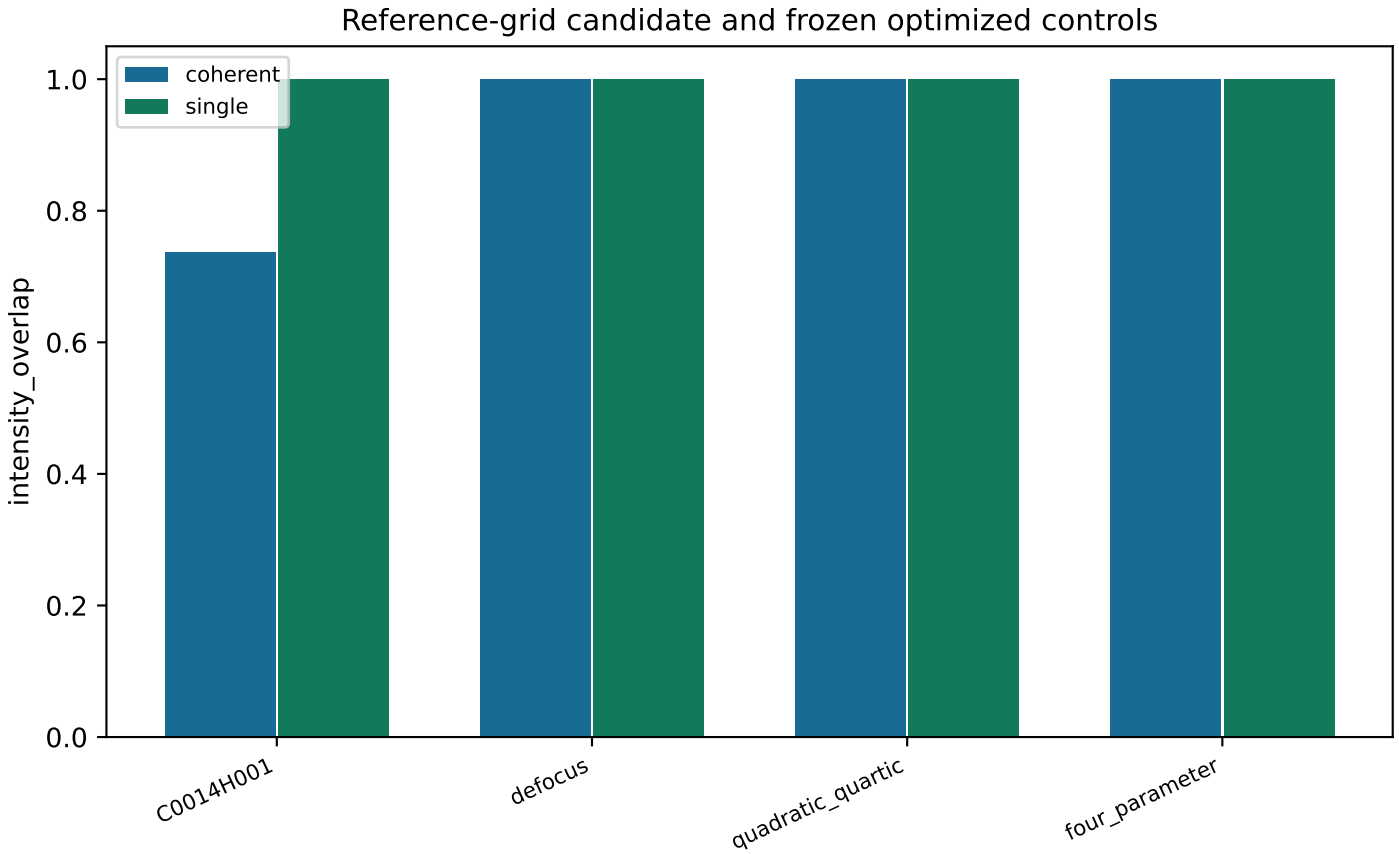
Evidence batch cycle_0014 (continued)

independent_observable_validation, reference_physics. Failed robustness probes: quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: True.

C0014H003: unconfirmed; reference effect 0.682966; numerical error floor 0.01. Failed confirmation gates: mesh_convergence, window_convergence, z_convergence, independent_observable_validation, reference_physics. Failed robustness probes: quantization, held_out_wavelength, noise_107, noise_109. Registered direction supported beyond error: False. Frozen optimized controls passed basic checks: False.

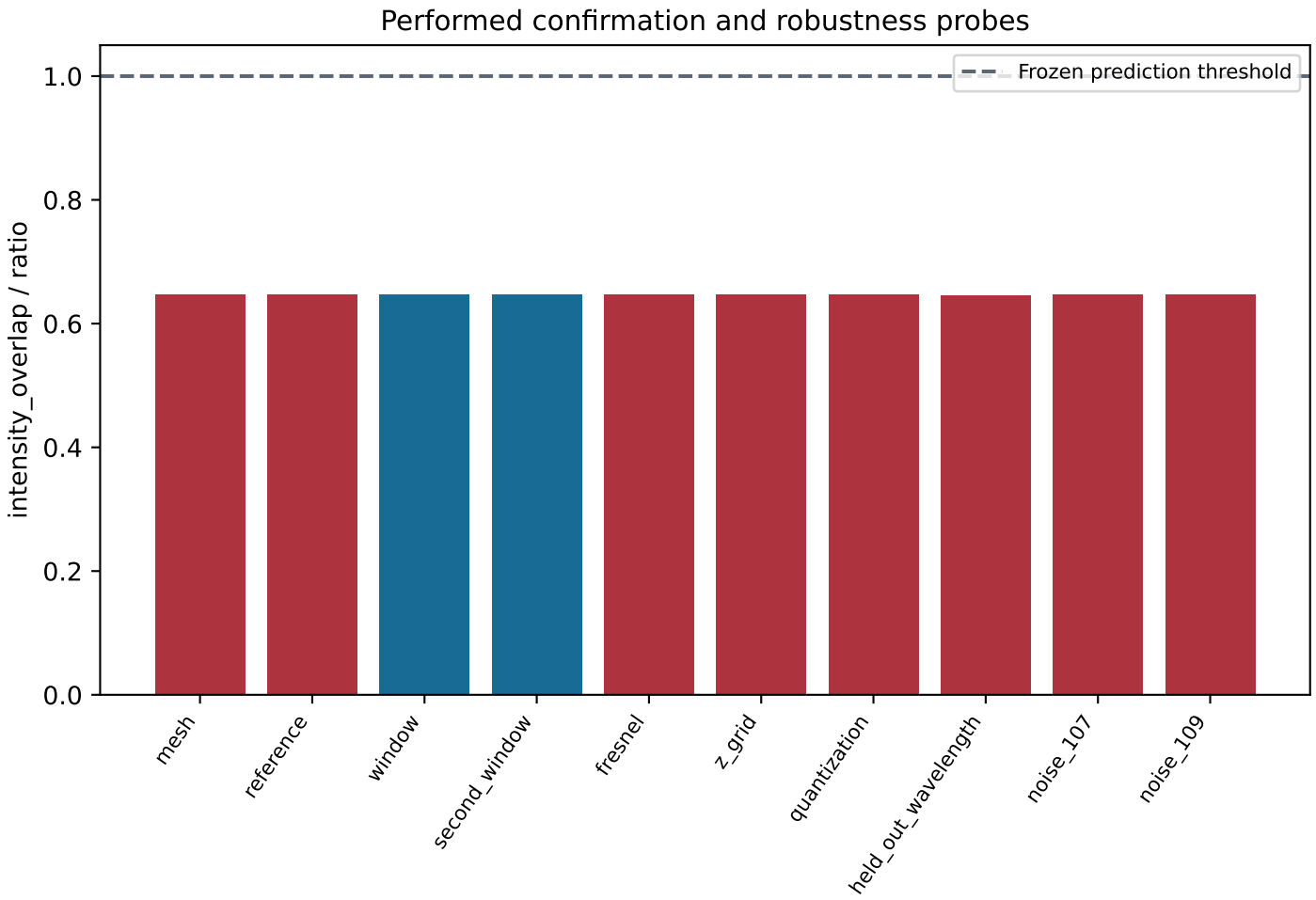
cycle_0014 / C0014H001

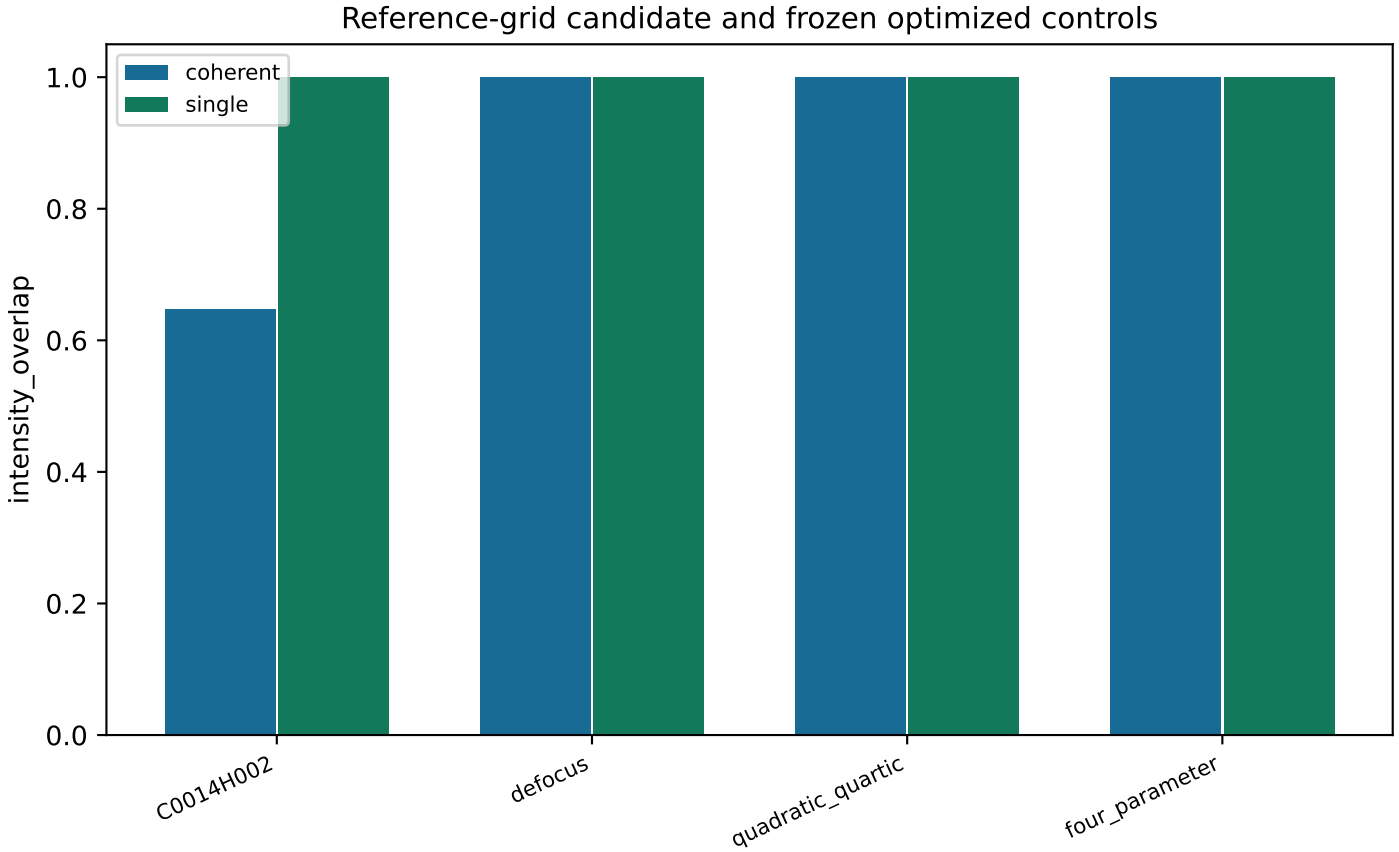




Red bars fail at least one numerical gate. Missing measurements are not zero.
Controls are re-evaluations of frozen parameters, not exhaustive optima or a novelty test.

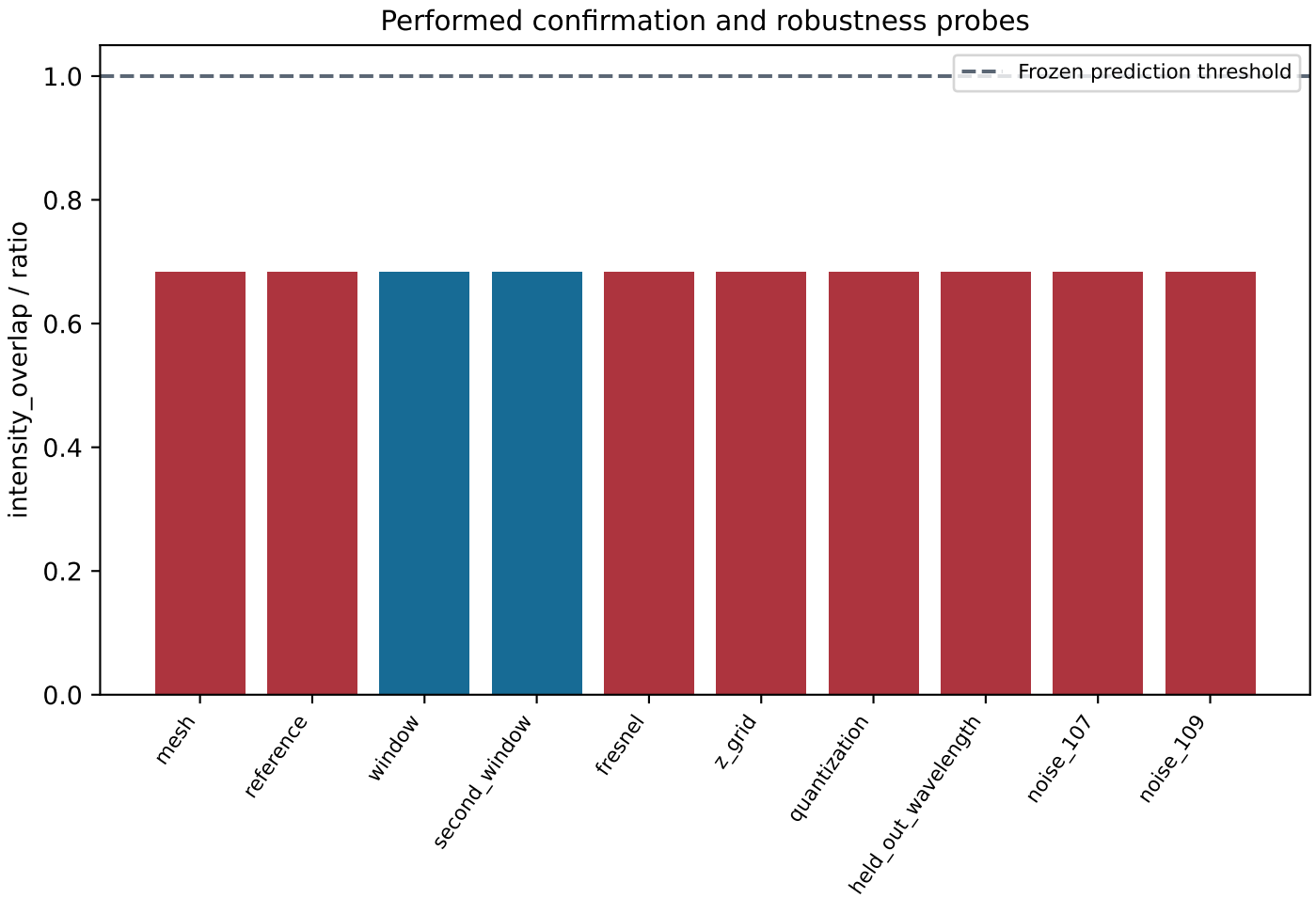
cycle_0014 / C0014H002

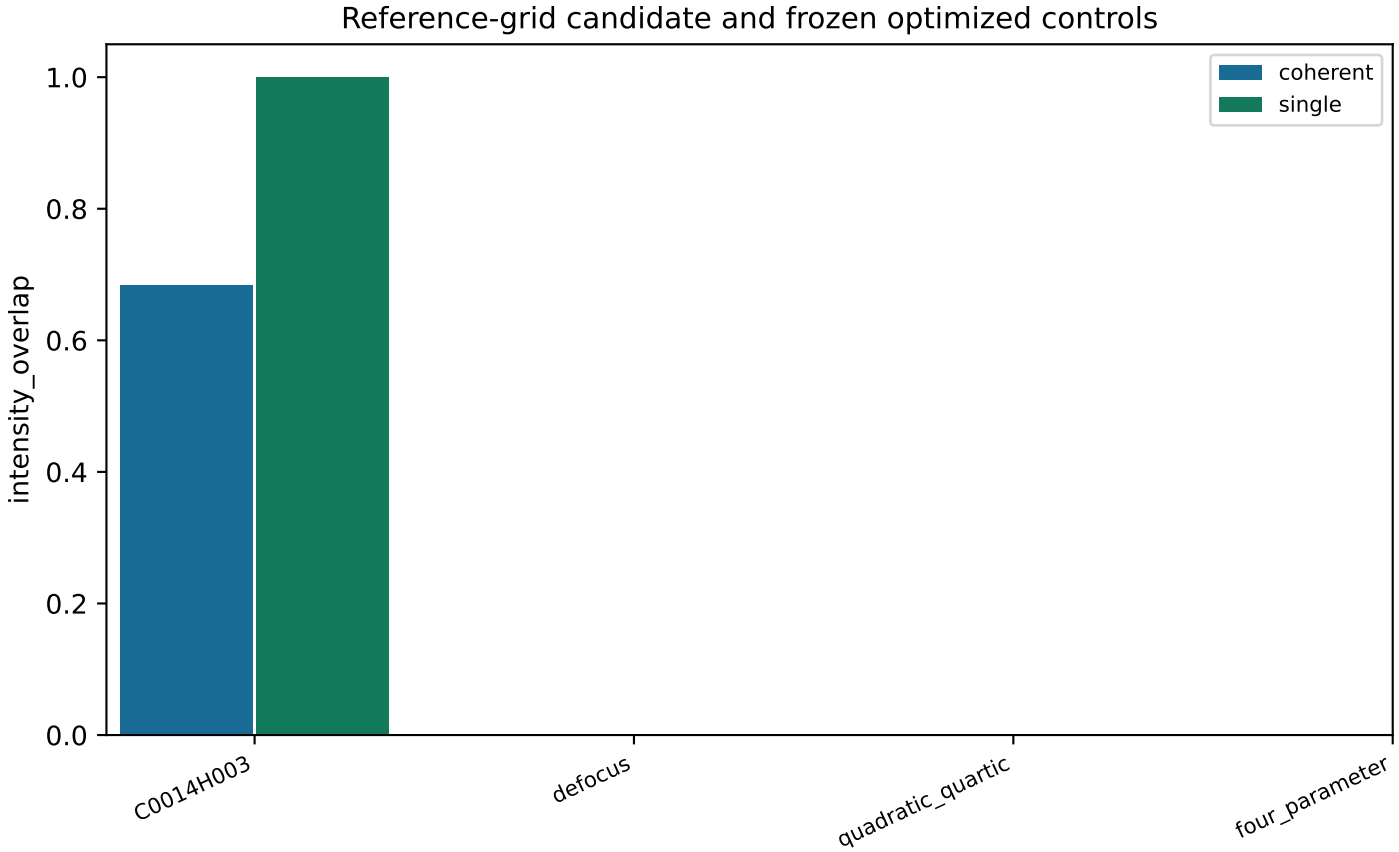




Red bars fail at least one numerical gate. Missing measurements are not zero.
Controls are re-evaluations of frozen parameters, not exhaustive optima or a novelty test.

cycle_0014 / C0014H003





Red bars fail at least one numerical gate. Missing measurements are not zero.
Controls are re-evaluations of frozen parameters, not exhaustive optima or a novelty test.

Physical interpretation and falsification

Coherent composition changes both local intensity and its equal-power normalization. If each constituent has power P, define input overlap S = integral(conj(E1)*E2)/P and the propagated core cross term C(z) = integral_core(conj(U1)*U2)/P. For amplitude weight w and phase phi, coherent core efficiency is [A(z)+w^2 B(z)+2w Re(exp(i phi) C(z))] / [1+w^2+2w Re(exp(i phi) S)]. The incoherent value is [A(z)+w^2 B(z)]/(1+w^2). The report statistic takes the registered median over z.

This familiar interference identity explains why changing phase can redistribute core power without creating optical power. It does not establish a Pearcey-specific mechanism. A mechanism study must isolate overlap and normalization, use matched Gaussian and asymmetric controls, and test frozen held-out settings. Opposite-phase negative effects can falsify an overly broad nonnegative prediction while remaining reproducible observations.

Known or invalid situations must remain explicit: a Gaussian and its half-turn copy can be identical up to global phase; odd-charge vortex fields can cancel exactly. Equal-power renormalization of a zero field is undefined. A transformed-versus-base comparison preserves the declared environment, including any lens. The explicit lens-versus-free-space comparison is required to test the same source with and without the lens.

Boundary tests matter for long Airy tails and ridge identification. A large apparent valley-filling effect that changes under window enlargement cannot support a stable quantitative claim. Spatial refinement alone is insufficient. The available windows and tolerance checks are finite diagnostics and do not prove an asymptotic convergence theorem.

Next discriminating experiment. Freeze a complete charge-by-phase or family-by-phase factorial design with balanced controls, a common observable and hold-out parameters. Execute it through the trusted structured interface, retain all failures and signs, then compare with specific prior results. Do not select a favorable core radius, propagation window, prediction direction or phase after inspecting the same data.

References, provenance and limitations

The following are source-library references with project-recorded provenance. This run does not revalidate their full text, and they do not constitute an exhaustive novelty search. No new citation or DOI is invented.

Accelerating finite energy Airy beams (2007). DOI: 10.1364/OL.32.000979. Source-library verification: Publisher metadata and original article first page inspected; solution checked against authors' 2012 chapter..

Self-accelerating Airy Beams: Generation, Control, and Applications (2012). DOI: 10.1007/978-1-4614-3538-9_1. Source-library verification: Author-hosted full chapter inspected, Eq. (1.6), Fig. 1.3; x0=100 micrometres, a=0.1, wavelength=0.5 micrometres, z=0,0.314,0.628,0.943,1.257 m..

Auto-focusing and self-healing of Pearcey beams (2012). DOI: 10.1364/OE.20.018955. Source-library verification: Publisher metadata verified. Canonical integral independently checked in Zannotti et al. 2017; no self-healing result imported..

Optical catastrophes of the swallowtail and butterfly beams (2017). DOI: 10.1088/1367-2630/aa6ecd. Source-library verification: Author-hosted full text inspected: section 2 Eqs. (1)-(2), section 5 and Fig. 4. Our Butterfly fixes a3=a4=0 and adds an explicit Gaussian envelope..

Orbital angular momentum of light and the transformation of Laguerre-Gaussian laser modes (1992). DOI: 10.1103/PhysRevA.45.8185. Source-library verification: Publisher metadata and abstract; implemented p=0 LG spatial form, no quantitative OAM claim..

Laser beams and resonators (1966). DOI: 10.1364/AO.5.001550. Source-library verification: Bibliographic metadata verified; Gaussian solver limit independently checked analytically..

Reproducibility. Each batch directory contains its frozen protocol, original measurements, a frozen validation protocol, persistent queue, reference/variant measurements, solver settings, per-hypothesis assessment and re-evaluated controls. The run evaluation journal records started, completed and failed attempts. The PDF is rebuilt deterministically from these records; JSON files remain authoritative.

Limitations. Scalar and paraxial models have finite applicability; high-angle vector and nonlinear effects are outside scope. Robustness probes are limited fixed tests, not population statistics. Source-library metadata is not an exhaustive technical prior-art review. Human scientific review remains necessary before a research or publication claim.